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Report No: PADHI01446

INTERNATIONAL DEVELOPMENT ASSOCIATION

PROJECT APPRAISAL DOCUMENT

ON A

PROPOSED GRANT

IN THE AMOUNT OF

SDR 56.1 MILLION (US\$76.1 MILLION EQUIVALENT)  
FROM THE GLOBAL AND REGIONAL OPPORTUNITIES WINDOW

AND

SDR 61.9 MILLION (US\$83.9 MILLION EQUIVALENT)

TO THE

FEDERAL REPUBLIC OF SOMALIA

FOR A

SOMALIA INFRASTRUCTURE DEVELOPMENT FOR ACCESS AND JOBS, PHASE 1

AS PHASE 1 OF THE MULTI-PHASE PROGRAMMATIC APPROACH  
SOMALIA INFRASTRUCTURE DEVELOPMENT FOR ACCESS AND JOBS PROGRAM

WITH AN OVERALL FINANCING ENVELOPE OF US\$500 MILLION EQUIVALENT

April 13, 2026

Transport Global Practice  
Urban Resilience and Land Global Practice  
Eastern and Southern Africa Region

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## CURRENCY EQUIVALENTS

(Exchange Rate Effective March 31, 2026)

Currency Unit =

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US\$1 = SDR 0.74

FISCAL YEAR

July 1 - June 30

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## ABBREVIATIONS AND ACRONYMS

|                 |                                                                |
|-----------------|----------------------------------------------------------------|
| ADS-B           | Automatic Dependent Surveillance–Broadcast                     |
| ARFF            | Aircraft Rescue and Fire Fighting                              |
| ATM             | Air Traffic Management                                         |
| BCR             | Benefit-Cost Ratio                                             |
| BRA             | Banadir Regional Administration                                |
| CBA             | Cost-Benefit Analysis                                          |
| CCDR            | Country Climate and Development Report                         |
| CNS             | Communication, Navigation and Surveillance                     |
| CO2             | Carbon Dioxide                                                 |
| CRI             | Corporate Results Indicator                                    |
| DA              | Designated Account                                             |
| DG              | Director General                                               |
| E&S             | Environmental and Social                                       |
| EAFD            | External Assistance Fiduciary Directorate                      |
| EIRR            | Economic Internal Rate of Return                               |
| ERA             | Ethiopian Roads Authority                                      |
| FCV             | Fragile, Conflict, and Violence                                |
| FGS             | Federal Government of Somalia                                  |
| FMS             | Federal Member State                                           |
| GBV             | Gender-Based Violence                                          |
| GHG             | Greenhouse Gas                                                 |
| GRS             | Grievance Redress Service                                      |
| ICAO            | International Civil Aviation Organization                      |
| IFC             | International Finance Corporation                              |
| IPF             | Investment Project Financing                                   |
| M&E             | Monitoring & Evaluation                                        |
| MIGA            | Multilateral Investment Guarantee Agency                       |
| MoF             | Ministry of Finance                                            |
| MoPWRH          | Ministry of Public Works, Reconstruction and Housing           |
| MoTCA           | Ministry of Transport and Civil Aviation                       |
| MPA             | Multi-phase Programmatic Approach                              |
| NBS             | Nature Based Solutions                                         |
| NPV             | Net Present Value                                              |
| O&M             | Operations and Maintenance                                     |
| PDO             | Project Development Objective                                  |
| PFM             | Public Financial Management                                    |
| PIU             | Project Implementation Unit                                    |
| PrDO            | Program Development Objective                                  |
| RAP             | Resettlement Action Plan                                       |
| SCAA            | Somalia Civil Aviation Authority                               |
| SEA/SH          | Sexual Exploitation and Abuse / Sexual Harassment              |
| SHIIP           | Somalia – Horn of Africa Infrastructure Integrated Project     |
| Stairs2Jobs MPA | Somalia Infrastructure Development for Access and Jobs Program |

|         |                                                   |
|---------|---------------------------------------------------|
| STEP    | Systematic Tracking of Exchanges in Procurement   |
| SURP-II | Somalia Urban Resilience Project Phase 2 (Nagaad) |
| ToC     | Theory of Change                                  |
| UN      | United Nations                                    |
| VSAT    | Very Small Aperture Terminals                     |



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**DATASHEET****BASIC INFORMATION**

|                                                          |                                                                                       |                                                          |  |
|----------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------|--|
| Project Beneficiary(ies)<br>Somalia, Federal Republic of | Operation Name<br><br>Somalia Infrastructure Development for Access and Jobs, Phase 1 |                                                          |  |
| Operation ID<br><br>P513127                              | Financing Instrument<br><br>Investment Project Financing (IPF)                        | Environmental and Social Risk Classification<br><br>High |  |

**Financing & Implementation Modalities**

|                                                                            |                                                                          |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> Multiphase Programmatic Approach (MPA) | <input type="checkbox"/> Contingent Emergency Response Component (CERC)  |
| <input type="checkbox"/> Series of Projects (SOP)                          | <input checked="" type="checkbox"/> Fragile State(s)                     |
| <input type="checkbox"/> Performance-Based Conditions (PBCs)               | <input type="checkbox"/> Small State(s)                                  |
| <input type="checkbox"/> Financial Intermediaries (FI)                     | <input type="checkbox"/> Fragile within a non-fragile Country            |
| <input type="checkbox"/> Project-Based Guarantee                           | <input checked="" type="checkbox"/> Conflict                             |
| <input type="checkbox"/> Deferred Drawdown                                 | <input type="checkbox"/> Responding to Natural or Man-made Disaster      |
| <input type="checkbox"/> Alternative Procurement Arrangements (APA)        | <input type="checkbox"/> Hands-on Expanded Implementation Support (HEIS) |

|                                       |                                      |                                              |
|---------------------------------------|--------------------------------------|----------------------------------------------|
| Expected Approval Date<br>05-May-2026 | Expected Closing Date<br>30-Jun-2031 | Expected Program Closing Date<br>26-Jun-2036 |
| Bank/IFC Collaboration<br>No          |                                      |                                              |

**MPA Program Development Objective**

to increase access to resilient infrastructure, employment, and services for Somalis in targeted areas.

**MPA FINANCING DATA (US\$, Millions)**

|                                |        |
|--------------------------------|--------|
| MPA Program Financing Envelope | 500.00 |
|--------------------------------|--------|

**Proposed Development Objective(s)**

to increase access to resilient infrastructure and infrastructure sector jobs, and improve the capacity of select institutions.

**Components**

| Component Name                                              | Cost (US\$)    |
|-------------------------------------------------------------|----------------|
| Resilient Infrastructure for Connectivity, Jobs, and Growth | 138,200,000.00 |
| Institutional Transformation and Human Capacity Building    | 5,800,000.00   |
| Sustainable Operations and Management                       | 8,500,000.00   |
| Infrastructure Implementation Coordination and Management   | 7,500,000.00   |

**Organizations**

Borrower/Recipient: Federal Republic of Somalia  
Implementing Agency: Ministry of Transport and Civil Aviation, Ministry of Public Works, Reconstruction and Housing, Ministry of Finance

**MPA FINANCING DETAILS (US\$, Millions)**

|                                       |        |
|---------------------------------------|--------|
| Board Approved MPA Financing Envelope | 0.00   |
| MPA Program Financing Envelope:       | 500.00 |
| of which Bank Financing (IBRD):       | 0.00   |
| of which Bank Financing (IDA):        | 500.00 |
| of which Other Financing sources:     | 0.00   |

**PROJECT FINANCING DATA (US\$, Millions)****Maximizing Finance for Development**

Is this an MFD-Enabling Project (MFD-EP)? No  
Is this project Private Capital Enabling (PCE)? Yes

**SUMMARY**

|                      |        |
|----------------------|--------|
| Total Operation Cost | 160.00 |
|----------------------|--------|



|                          |               |
|--------------------------|---------------|
| <b>Total Financing</b>   | <b>160.00</b> |
| <b>of which IBRD/IDA</b> | <b>160.00</b> |
| <b>Financing Gap</b>     | <b>0.00</b>   |

**DETAILS****World Bank Group Financing**

|                                             |        |
|---------------------------------------------|--------|
| International Development Association (IDA) | 160.00 |
| of which IDA Recommitted                    | 30.00  |
| IDA Grant                                   | 160.00 |

**IDA Resources (US\$, Millions)**

|                                                 | Credit Amount | Grant Amount  | SML Amount  | Guarantee Amount | Total Amount  |
|-------------------------------------------------|---------------|---------------|-------------|------------------|---------------|
| National Performance-Based Allocations (PBA)    | 0.00          | 83.90         | 0.00        | 0.00             | 83.90         |
| Global and Regional Opportunities Window (GROW) | 0.00          | 76.10         | 0.00        | 0.00             | 76.10         |
| <b>Total</b>                                    | <b>0.00</b>   | <b>160.00</b> | <b>0.00</b> | <b>0.00</b>      | <b>160.00</b> |

**Expected Disbursements (US\$, Millions)**

| WB Fiscal Year | 2026 | 2027  | 2028  | 2029   | 2030   | 2031   |
|----------------|------|-------|-------|--------|--------|--------|
| Annual         | 1.00 | 20.00 | 30.00 | 50.00  | 40.00  | 19.00  |
| Cumulative     | 1.00 | 21.00 | 51.00 | 101.00 | 141.00 | 160.00 |

**PRACTICE AREA(S)****Practice Area (Lead)****Contributing Practice Areas**



Transport

Urban, Resilience and Land

## CLIMATE

### Climate Change and Disaster Screening

Yes, it has been screened and the results are discussed in the Operation Document

## SYSTEMATIC OPERATIONS RISK- RATING TOOL (SORT)

| Risk Category                                                   | Rating        |
|-----------------------------------------------------------------|---------------|
| 1. Political and Governance                                     | ● High        |
| 2. Macroeconomic                                                | ● Substantial |
| 3. Sector Strategies and Policies                               | ● Substantial |
| 4. Technical Design of Project or Program                       | ● Substantial |
| 5. Institutional Capacity for Implementation and Sustainability | ● Substantial |
| 6. Fiduciary                                                    | ● High        |
| 7. Environment and Social                                       | ● High        |
| 8. Stakeholders                                                 | ● High        |
| 9. Other                                                        | ● High        |
| 10. Overall                                                     | ● High        |
| <b>Overall MPA Program Risk</b>                                 | ● High        |

## POLICY COMPLIANCE

### Policy

Does the project depart from the CPF in content or in other significant respects?

☐ Yes ☒ No

Does the project require any waivers of Bank policies?

☐ Yes ☒ No



## ENVIRONMENTAL AND SOCIAL

### Environmental and Social Standards Relevance Given its Context at the Time of Appraisal

| E & S Standards                                                                                      | Relevance              |
|------------------------------------------------------------------------------------------------------|------------------------|
| ESS 1: Assessment and Management of Environmental and Social Risks and Impacts                       | Relevant               |
| ESS 10: Stakeholder Engagement and Information Disclosure                                            | Relevant               |
| ESS 2: Labor and Working Conditions                                                                  | Relevant               |
| ESS 3: Resource Efficiency and Pollution Prevention and Management                                   | Relevant               |
| ESS 4: Community Health and Safety                                                                   | Relevant               |
| ESS 5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement                       | Relevant               |
| ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources              | Relevant               |
| ESS 7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities | Not Currently Relevant |
| ESS 8: Cultural Heritage                                                                             | Relevant               |
| ESS 9: Financial Intermediaries                                                                      | Not Currently Relevant |

NOTE: For further information regarding the World Bank's due diligence assessment of the Project's potential environmental and social risks and impacts, please refer to the Project's Appraisal Environmental and Social Review Summary (ESRS).

## LEGAL

### Legal Covenants

#### Sections and Description

Schedule 2, Section I.A.1(a) - Establish, not later than ninety (90) days after the Effective Date, and thereafter maintain throughout the period of Project implementation, a ministerial committee ("Ministerial Committee"): (i) chaired by the Minister of Finance, and comprising of high level representatives, including from the MoPWRH, MoTCA, and Office of the Prime Minister; and (ii) vested with such powers, functions and competencies, acceptable to the Association and set forth in the Project Operations Manual ("Project Operations Manual" or "POM"), as shall be required to provide strategic direction for the project.

Schedule 2, Section I.A.1(b) - Establish, not later than ninety (90) days after the Effective Date, and thereafter maintain throughout the period of Project implementation, a technical committee ("Technical Committee"): (i) chaired by the DG of the Ministry of Finance, and comprising of, inter alia, directors-general of MoTCA and MoPWRH, and representation from Federal Member States (where applicable); and (ii) vested with such powers, functions and competencies, acceptable to the Association and set forth in the POM, as shall be required to provide technical oversight and day to day governance of the Project.

Schedule 2, Section I.A.1(d) - Establish not later than ninety (90) days after the Effective Date, and thereafter maintain throughout the period of Project implementation, an MPA Program coordination Unit within MoF ("MoF-PCU"): (i) led



by a coordinator, and assisted by competent, experienced and qualified staff, including a monitoring & evaluation/communications specialist, a procurement specialist, a financial management specialist, and such other specialists as may be required, under terms of reference acceptable to the Association; and (ii) vested with such powers, resources, functions and competencies, acceptable to the Association and set forth in the POM, as shall be required to coordinate the MPA Program and carry out Part 2(c) and its relevant activities under Part 4 of the Project.

#### Conditions

| Type          | Citation                       | Description                                                                                                                                                                                                                                               | Financing Source |
|---------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Effectiveness | Article IV, Section 4.01(a)    | The Recipient has prepared and adopted the Project Operations Manual (POM) in form and substance satisfactory to the Association.                                                                                                                         | IBRD/IDA         |
| Effectiveness | Article IV, Section 4.01(b)    | The Recipient has designated a Coordinating Environmental and Social Specialist within the Ministry of Public Works, Reconstruction, and Housing (MoPWRH) with qualifications, experience, and under terms of reference, satisfactory to the Association. | IBRD/IDA         |
| Disbursement  | Schedule 2, Section III.B.1(b) | Under Category (2)(a) unless and until the Recipient has recruited to MoPWRH-PIU, a dedicated health and safety expert, with experience, qualifications and under terms of reference satisfactory to the Association.                                     | IBRD/IDA         |
| Disbursement  | Schedule 2, Section III.B.1(c) | Under Category (3)(a) unless and until the Recipient has established MoTCA-PIU, with composition and resources satisfactory to the Association, and has recruited to the MoTCA-                                                                           | IBRD/IDA         |



|              |                                |                                                                                                                                                                                                                                                                                                                       |          |
|--------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|              |                                | PIU: (i) a coordinator, (ii) procurement specialist, (iii) environmental, health and safety specialist, (iv) social specialist, and (v) financial management specialist, all with qualifications, experience, and under terms of reference satisfactory to the Association.                                           |          |
| Disbursement | Schedule 2, Section III.B.1(d) | Under Categories (2)(a) and (3)(a) unless and until the Recipient has prepared, consulted upon, and disclosed the relevant Environmental Social Impact Assessments (ESIA) and the Environmental and Social Management Plan (ESMP), in form and substance satisfactory to the Association                              | IBRD/IDA |
| Disbursement | Schedule 2, Section III.B.1(d) | For Eligible Expenditures under Category (4), until and unless the Association has notified the Recipient that the conditions set forth in Section 5.15 (a) of the General Conditions have been fulfilled                                                                                                             | IBRD/IDA |
| Disbursement | Schedule 2, Section III.B.1(e) | For Complementary Financing for the Cat DDO under Category (5), until and unless: (i) the Recipient has furnished to the Association a request to reallocate and thereafter withdraw all or part of the Unwithdrawn Credit Balance for the Complementary Financing for the Cat DDO, and such notice specifies the Cat | IBRD/IDA |



|  |  |                                                                                                                                                                                                                                                                                                  |  |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  |  | DDO Legal Agreement; and (ii) the Association has accepted said request and notified the Recipient thereof, and is satisfied, based on evidence satisfactory to it, that the conditions precedent to withdrawal of the financing provided under the Cat DDO Legal Agreement have been fulfilled. |  |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|



## I. STRATEGIC CONTEXT

### A. Country Context

1. **The Federal Republic of Somalia continues to make notable progress in consolidating its federal governance system, strengthening public institutions, and fostering the inclusive, private sector-led development critical for job creation.** Despite these advances, structural constraints persist, including an unfinished political settlement, fragmented governance, and high levels of insecurity which continue to weigh on confidence and investment. Private sector activity remains concentrated in a few urban centers where basic infrastructure lowers operating costs; yet urban poverty remains substantial (46 percent of urban residents live below the national poverty line).<sup>1</sup> Somalia's main cities nonetheless represent a key opportunity for resilience and growth: roughly 90 percent of the urban population in cities of 20,000 or more is concentrated along six strategic corridors.

2. **The development context remains shaped by high poverty, recurrent shocks, and escalating climate risks.** Over half of the population live below the poverty line, with food insecurity magnifying their vulnerability with 6.5 million people at crisis level food insecurity.<sup>2</sup> Climate change impacts including recurrent droughts, floods, extreme heat, and rainfall variability are placing mounting pressure on municipal infrastructure and services, contributing to the expansion of informal settlements and heightened exposure to floods and droughts. Transport infrastructure is particularly vulnerable to flooding, heat stress, and storm events that can disrupt corridor connectivity and aviation safety and operations.<sup>3</sup> Reflecting these pressures, economic growth has moderated since 2024: real Gross Domestic Product (GDP) growth slowed to an estimated 3.0 percent in 2025 and is projected at 2.8 percent in 2026, as sharp aid cuts, worsening drought conditions, and external spillovers weigh on consumption, rural incomes, and exports. Significant risks remain, given uncertainty around aid, climate shocks, security conditions, and global price volatility.<sup>4</sup>

### B. Sectoral and Institutional Context

3. **Foundational infrastructure deficits remain a binding constraint to growth, resilience, and construction-led recovery within and between cities.** Construction is a critical entry point for jobs and post conflict recovery, yet its economic impact is constrained by informality, limited scale, and weak firm capabilities, constraints that are exacerbated by poor infrastructure.<sup>5</sup> Limited road coverage and poor maintenance restrict mobility and market access. An ongoing Mogadishu flood risk assessment estimates US\$120 million in average annual losses; 61 percent of households reporting income losses driven by the inability of workers to reach workplaces, and disruptions to supply chains.<sup>6</sup> Improved drainage and transport reliability will reduce flood risk, improve reliable access, lower construction costs, enable year-round work, and support sustained private sector growth and job creation.

4. **Economic fragmentation extends beyond urban centers to rural urban linkages and regional trade, limiting the benefits of agglomeration.** Somalia's 3,300 kilometers coastline and major seaports offer potential as a regional trade hub for the Horn of Africa, linking Africa, Asia, Europe, and the Middle East. Yet fragmented logistics networks, security constraints, and weak infrastructure continue to drive high transport costs even over short distances, contributing to price disparities and limiting market integration. Corridor and urban road investments reduce the cost of moving goods from production centers to markets, while drainage and urban infrastructure improvements ensure that receiving and distribution points in primary cities remain functional year-round. Air connectivity directly addresses

<sup>1</sup> *Somalia Poverty and Equity Assessment*, based on the Somalia Integrated Household Budget Survey (SIHBS) 2022. Washington, DC: World Bank.

<sup>2</sup> *Somalia: IPC Acute Food Insecurity and Acute Malnutrition Analysis (January–June 2026)*. February 24, 2026. ReliefWeb.

<sup>3</sup> World Bank. 2026. Country Climate and Development Report (English). Washington, D.C. : World Bank Group.

<sup>4</sup> World Bank. Somalia Macro-Poverty Outlook. October 2025

<sup>5</sup> World Bank (2025). *Private Sector in the Federal Republic of Somalia: Composition, Performance, and Constraints to Growth*.

<sup>6</sup> Ongoing World Bank Flood Risk Assessment study in Mogadishu and Hargeisa, 2026



this constraint by providing reliable access between primary cities and underserved regions. It also protects humanitarian logistics capacity while enabling commercial air services on which perishable exports depend.

**5. Addressing spatial and economic fragmentation requires a strategic approach to reducing functional economic distances between cities and their hinterlands.** The concept of “basins of integration” is central to this strategy - geographically connected economic regions, often structured around key transport and trade corridors, where multisectoral investments are synchronized to drive agglomeration economies. Developing these basins demands coordinated upgrades to road, air transport, energy, and digital connectivity, alongside trade facilitation and institutional strengthening. A Horn of Africa infrastructure impact study estimates that such coordinated interventions could yield a 12 percent increase in the employment share of services and a 6.3 percent rise in real incomes in Somalia.<sup>7</sup> Interventions in such basins can also unlock opportunities for small and medium-sized enterprises (SMEs), microenterprises, and a range of service providers both directly, through transport and logistics, and indirectly, by supporting ancillary activities like supply chain management, veterinary services, and finance.

**6. Infrastructure governance and institutional frameworks in the Federal Republic of Somalia are evolving, with increasing, though uneven capacity across sectoral ministries and agencies.** Overlapping mandates across several ministries, including the Ministry of Public Works, Reconstruction and Housing (MoPWRH), and the Ministry of Transport and Civil Aviation (MoTCA), pose hurdles to integrated planning and service delivery. Financial sustainability is an ongoing challenge, as infrastructure development relies heavily on donor support, with limited mechanisms for maintenance and reinvestment. The potential for revenue generation, particularly in aviation, is strong but hampered by insufficient financial autonomy and lack of transparent reinvestment frameworks.

### C. Relevance to Higher Level Objectives

**7. Strategic sectoral direction is anchored in the Federal Republic of Somalia’s National Transformation Plan, which prioritizes resilient service delivery, inclusive private sector growth, and improved infrastructure quality.** The proposed program invests in foundational infrastructure while also improving governance, policies and skills particularly in the transport, urban and resilience sectors; thus, strategically aligning with the World Bank’s Jobs Strategy.<sup>8</sup> The Project is consistent with the World Bank Group Somalia Country Partnership Framework FY24-28 discussed by the Board on February 29, 2024 (Report No: 187254-SO). It implements the recommendations of the 2025 Somalia Country Climate and Development Report (CCDR), which identifies access to climate resilient infrastructure and flood risk management as critical to safeguarding growth and jobs. It is also aligned with the WBG’s Refreshed Fragile, Conflict and Violence (FCV) Strategy, which highlights how investments in foundational infrastructure are not just crucial to creating jobs and economic opportunities, but also for reducing FCV risks over time.<sup>9</sup> By supporting the Government to deliver tangible results to communities, the project will also contribute to strengthening the relationship between the state and its citizens, in line with the World Bank’s 2023 Somalia Risk and Resilience Assessment recommendations. The program is Paris aligned, integrating adaptation and mitigation measures embedded in national climate commitments, including the Nationally Determined Contributions and National Adaptation Plan Framework, and addresses binding constraints highlighted in core diagnostics.

### D. Multiphase Programmatic Approach (MPA)

#### i) Rationale for Using MPA

**8. In Somalia, the MPA brings multiple benefits as compared to a series of standalone projects:**

<sup>7</sup> Matias Herrera Dappe and Mathilde Lebrand. “Infrastructure and Structural Change in the Horn of Africa”. World Bank. 2021

<sup>8</sup> World Bank Group. (2025). *Foundations for growth and jobs* (Development Committee Paper No. DC2025-0004). World Bank.

<sup>9</sup> See: [https://www.worldbank.org/content/worldbankgroup/consultations/en/detail/wbg-fcv-strat-25-30.html?cq\\_ck=1770233346123](https://www.worldbank.org/content/worldbankgroup/consultations/en/detail/wbg-fcv-strat-25-30.html?cq_ck=1770233346123)



- **It provides a risk-based approach, enabling adaptive implementation, flexibility, and incremental capacity building to manage shocks.** It is informed by analytical work on delivering services in highly insecure areas,<sup>10</sup> emphasizing phased, risk informed, and institutionally grounded investments. In Somalia's fragile and dynamic FCV context, the MPA supports iterative scaling, learning across phases, and course correction as risks and opportunities evolve while keeping focus on the Program's outcomes, as laid out in the Theory of Change.
- **It is designed to unlock spatial synergies and deliver impact at scale through a coordinated, resilient investment program anchored in a learning agenda.** The phased approach allows piloting and scaling, demonstrating results and building delivery capacity before expansion into more complex basins. Consolidating investments also support targeted, conflict-sensitive implementation.
- **It sequences public investment to create supportive conditions for the private sector.** Investments in regulatory frameworks and capacities in infrastructure sectors will enable predictable, long-term and risk informed private commitments. Engagement platforms can help signal where private capital can follow public investments, and the MPA's staged structure supports predictability while remaining responsive to emerging opportunities.
- **It creates a framework for longer-term planning with clear criteria for subsequent phases while managing annual financing cycles and ongoing fundraising needs.** Somalia's uncertain environment manifests itself in the current short-term planning cycles and a piecemeal approach to financing. The MPA enables multi-year and multi-sectoral investments, as well as iterative reforms to strengthen asset sustainability and sector capacity, clarifying longer-term actions and financing needs through defined outcomes, metrics, and shared pipeline.
- **It leverages the World Bank Group's experience and convening power to attract other financing.** It builds on ongoing investments and cross-sector analysis to address bottlenecks, while using existing institutional and project arrangements to maximize readiness. It also helps align and potentially crowd in partner financing, including bilateral and multilateral development partners, and links with International Finance Corporation (IFC) and Multilateral Investment Guarantee Agency (MIGA) programs.

## ii) Program Results Chain

9. **The Somalia Infrastructure Development for Access and Jobs Program (Stairs2Jobs MPA) is intended to support the high-level objective of increased access to resilient infrastructure, employment and services for Somalis in targeted economic basins.** It is intended to address key challenges facing the country through a structured and iterative approach. The planned 10-year MPA will be structured around three core pillars:

- **Pillar 1 – Enabling Resilient Infrastructure for Access to Jobs and Growth.** This pillar supports climate- and disaster-resilient infrastructure to unlock Somalia's basins of integration by (i) strengthening key economic nodes through resilient investments, improved services, and job creation, and by (ii) connecting these nodes via resilient transport links to enhance supply chains, market access, and inclusion in fragile areas. Investments cover basic urban services; economic corridors with aligned energy and digital infrastructure and trade facilitation; airport and air transport services; and disaster risk management. All investments apply climate-resilient design standards to address flooding, extreme heat, and weather variability. The pillar prioritizes targeted, spatially aligned investments to build an infrastructure foundation for jobs, informed by regular FCV analysis to manage conflict and political economy risks and support effective implementation.
- **Pillar 2 – Institutional Transformation and Capacity Building.** This pillar will support the Government to strengthen governance, support business enabling policies and a predictable regulatory environment, and deliver resilient infrastructure for critical institutions. Support will focus on enhancing policy frameworks, regulatory systems, and coordination mechanisms across the Federal Government of Somalia (FGS), Federal Member States (FMS), local governments, and the private sector to ensure infrastructure investments are risk-informed, climate-

<sup>10</sup> World Bank. *Delivering in Highly Insecure Areas in Somalia: Country and Global Lessons to Inform the World Bank's Engagement*. Advisory Services and Analytics, March 2025 (Official Use Only).

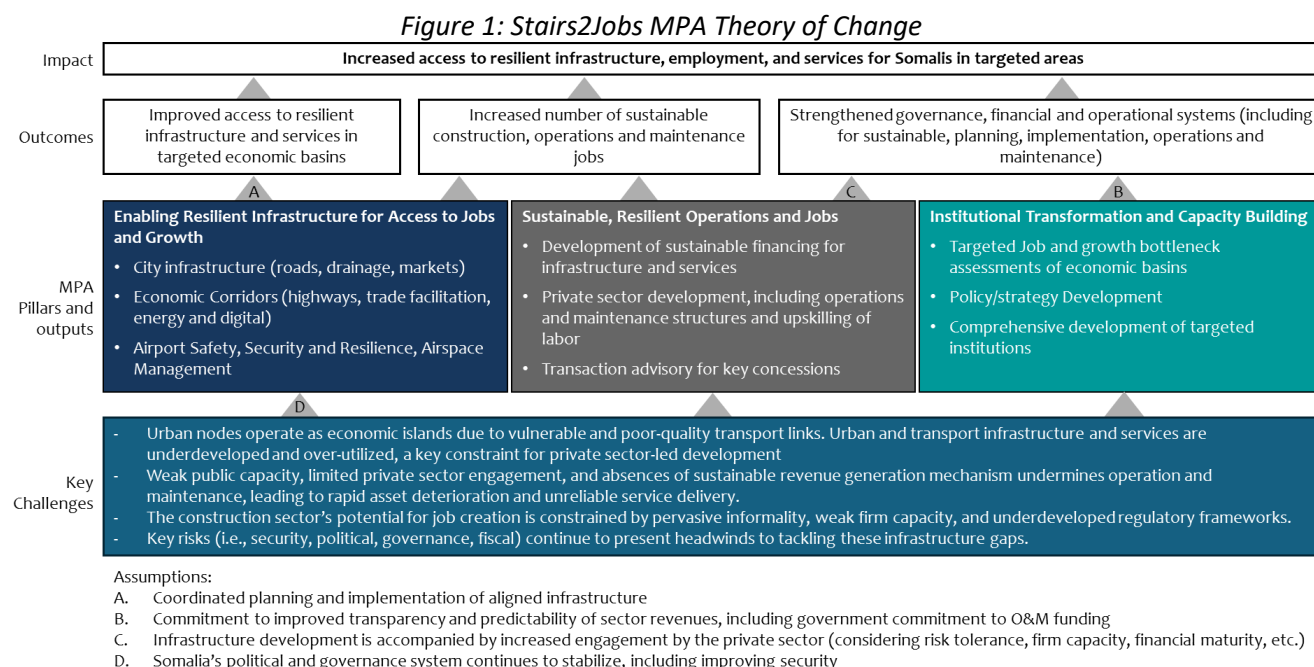


resilient, and aligned across sectors and geographies. Emphasis will be given to establishing and strengthening the licensing and regulatory functions needed to increase the availability of highly skilled professionals, such as an engineers' board. This includes supporting capacity building for planning, procurement, and asset management, including the integration of disaster risk management and climate resilience considerations.

- **Pillar 3 – Sustainable and resilient operations.** This pillar will support the development of organizational frameworks for the long-term sustainability and resilience of infrastructure assets. Interventions will include (i) transparent and sustainable financing mechanisms for key infrastructure sectors (such as overflight revenue, airport landing fees, fuel levies, road funds, user fees, and own-source revenue); (ii) models for private sector driven sustainable service delivery and the operations and maintenance of infrastructure assets; and (iii) transaction advisory services to support major concessions and crowd-in private capital.

10. **The Program is intended to provide benefits to three groups, (i) direct users of improved infrastructure, (ii) those who benefit from new or improved jobs, and (iii) those who benefit from the economic growth generated by such investments.** Direct users include those who drive on improved roads, travel through improved airports and airspace, or benefit from improved urban services. The program's support for job creation includes the significant number of construction jobs, as well as the longer-term jobs related to maintenance and operations of assets. It will also support upskilling and licensing of key professionals, such as engineers and air navigation services, unlocking practitioners of these professions' ability to find and retain better jobs. Finally, indirect beneficiaries include residents of targeted basins who are expected to benefit from the program's contribution to overall growth and improved resilience. By reducing functional economic distances via preventing flood-related disruption in urban cores and more reliable intercity connectivity, the Program is expected to expand the effective labor catchment and market reach of targeted cities, while enabling sustained private sector activity in areas constrained by infrastructure unreliability. Where surface transport remains constrained by security and infrastructure gaps, aviation provides the only reliable year-round connectivity between Somalia's major cities and to regional markets, making improvements a binding condition for market access, time-sensitive exports and humanitarian surge capacity.

11. **Figure 1 below describes the Theory of Change (ToC) of the Program.** It summarizes the points above, identifying the key problems the program is addressing, articulating the causal links between activities, outputs and outcomes, as well as highlighting the longer-term outcome to which the program is contributing.





### iii) Program Development Objective

12. **The Program Development Objective (PrDO)** is to increase access to resilient infrastructure, employment and services for Somalis in targeted areas.

13. **Program results indicators include:**

- People that benefit from improved access to sustainable transport infrastructure and services (CRI)
- Beneficiaries with enhanced resilience to climate risk (CRI)
- More and better-paid jobs created and enabled (CRI)

### iv) Program Framework

14. **The scope and timing of the future phases will depend on the availability of funding, the evolving FCV risk context, and continuous learning.** Future MPA phases will be guided by a government-led, transparent, and criteria-based sectoral and geographic prioritization framework, with a strong emphasis on readiness, as detailed under Subcomponent 2.C. The number of phases could be increased or reduced, as required, based on funding availability, emerging priorities, and implementation progress across components and activities.

15. **The first phase will prioritize basic access, climate resilience, and resource mobilization, while strengthening institutional capacity and establishing the core systems required to accelerate follow on investments.** Activities to be included in Phase 1 are intended to achieve three interconnected goals: (i) delivering early success to build stakeholder buy in, political momentum, and crowd in other financing; (ii) setting a foundation for future investments through institutional development and hands-on capacity building, including the integration of climate and disaster risk considerations, particularly in the area of flood risk management; and (iii) facilitating reliable, safe, and resilient access throughout the country to reduce implementation risks and costs for future phases.

16. **Building on the basic access provided through initial investments, future phases will deliver targeted support for the country's growing economic basins, adopting an integrated and tailored approach to local economic drivers and bottlenecks.** Subsequent phases will focus on spatially coordinated investments in key sectors. Investments will seek to match ambition with feasibility given the dynamic FCV risk context, and incorporate risk informed planning, resilient design standards, and, where feasible, nature-based solutions (NBS) to address flooding, drought, and other climate induced stresses. The rollout sequence of these phases will be determined by a readiness criterion, considering factors such as government priority and state presence, security, clear ownership of land, and the preparedness and feasibility of the included investments. To safeguard assets and consolidate development gains, disaster risk management and emergency preparedness measures will be mainstreamed across all interventions and phases.

17. **Phase 2 will scale up priority investments initiated under Phase 1, with a focus on those activities which have progressed furthest in implementation** (i.e. extension of the Mogadishu drainage system, the construction and upgrading of urban roads and selected economic corridors, and the improvement of additional airstrips to enhance safe, reliable access and strengthen national and regional connectivity).

18. **Phase 3 will replicate successful interventions from previous phases in additional locations.** It will prioritize reconstruction of national road infrastructure and urban investments within the country, learning from earlier interventions as well as the evolving security environment across the different basins. The corridors will be selected from a prioritized short list under design through an ongoing project (Somalia Horn of Africa Infrastructure Integrated Project -SHIIP, P173119). If institutional capacity building activities have progressed under prior phases, it will support further strengthening.

19. **Phase 4 will be the completion phase, with activities primarily concentrated in pillar 3.** It will support maintenance of critical infrastructure assets and institutionalization of operations and maintenance structures. In addition, it will be used to define future infrastructure interventions.



Table 1: Stairs2Jobs MPA Program Framework

| Phase #                           | Operation ID | Sequential or Simultaneous | Phase's Proposed DO*                                                                                                                         | IPF or PforR | Estimated IBRD Amount (\$ million) | Estimated IDA Amount (\$ million) | Estimated Other Amount (\$ million) | Estimated Approval Date | Estimated Environmental & Social Risk Rating |
|-----------------------------------|--------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------|-----------------------------------|-------------------------------------|-------------------------|----------------------------------------------|
| 1                                 | P513127      | Simultaneous               | To increase access to resilient infrastructure and infrastructure sector jobs, and improve the capacity of select institutions               | IPF          | 0.00                               | 160.00                            | 0.00                                | 5/5/2026                | High                                         |
| 2                                 | P515865      | Simultaneous               |                                                                                                                                              | IPF          | 0.00                               | 100.00                            | 0.00                                | FY28                    | High                                         |
| 3                                 | P515867      | Simultaneous               | To expand access to resilient corridor infrastructure and urban services, and to improve the capacity of relevant Ministries and Authorities | IPF          | 0.00                               | 190.00                            | 0.00                                | FY29                    | High                                         |
| 4                                 | P515868      | Simultaneous               | To enhance sustainable operations and management programs for core infrastructure assets                                                     | IPF          | 0.00                               | 50.00                             | 0.00                                | FY31                    | Substantial                                  |
| Total                             |              |                            |                                                                                                                                              |              | 0.00                               | 500.00                            | 0.00                                |                         |                                              |
| Total Program Envelope            |              |                            |                                                                                                                                              |              | 500.00                             |                                   |                                     |                         |                                              |
| Board Approved Financing Envelope |              |                            |                                                                                                                                              |              | 0.00                               |                                   |                                     |                         |                                              |



## v) Learning Agenda

20. Recognizing Somalia's complex and fragile context, as well as the mixed track record of infrastructure development and private sector participation, implementation of the MPA will rely on a robust learning agenda focused on four areas: (i) implementation of infrastructure at scale, (ii) iterative institutional development, (iii) infrastructure planning and prioritization, and (iv) FCV risk mitigation. Addressing them will be supported through spatially targeted, locally informed assessments, coordinated through the Livable Cities and Disaster Resilience platform (P507709), to strengthen alignment, sequencing, and cross sector coherence in future phases. These areas reflect the fundamental challenges that have proven difficult during previous projects, and which are likely to potentially present hurdles to achieving the MPA's Program Development Objective. The Learning Agenda is further developed in Table 2.

*Table 2: Learning Agenda*

| Key Learning Gaps                                 | Priority Learning Questions                                                                                                                                                                                                  | Learning Methodology                                                                                                                    | Institutional Responsibility                                                                                | Financing & Phase Linkages                                                                                     |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <i>Implementation of infrastructure at scale</i>  | Which delivery and contracting models deliver acceptable cost, time, quality, and Environmental and Social (E&S) outcomes in FCV settings? At what point can functions transition to civil servants without increasing risk? | Comparative scorecards across Phase 1 investments; structured supervision debriefs; piloting gradual integration of government systems. | MoPWRH, MoTCA (technical); MoF MPA Coordination Unit (synthesis); Technical Steering Committee (decisions). | Component 4 and embedded implementation costs; informs delivery models and supervision approaches for Phase 2. |
| <i>Iterative institutional development</i>        | Which institutional functions must be operational before asset scale-up? What milestones justify transition from design to operationalization?                                                                               | Phase-based institutional diagnostics; tracking legal, staffing, financing and early performance milestones.                            | Sector ministries with MoF consolidation and Steering Committee validation.                                 | Pillar 2 financing; Phase 2 support conditional on Phase 1 institutional milestones.                           |
| <i>Infrastructure planning and prioritization</i> | Which investments deliver highest resilience and jobs returns per dollar? Which reforms translate into funded Operations and maintenance (O&M)?                                                                              | Economic basin diagnostics; criteria-based prioritization framework; ex-post validation using Phase 1 outcomes.                         | MoF (Sub-component 2.C) with sector ministry inputs.                                                        | Sub-component 2.C and Pillars 2–3; directly informs Phase 2 sequencing and geographic expansion.               |
| <i>FCV risk mitigation</i>                        | Which mitigation measures reduce disruption, delays, and cost overruns? How should readiness criteria adjust as risks evolve?                                                                                                | Periodic FCV risk reviews; cross-site comparisons; integration into procurement and supervision.                                        | MoF coordination with implementing ministries and Bank FCV Risk Platform.                                   | Component 4 supervision resources; informs geographic sequencing and procurement approaches in later phases.   |



## II. PROJECT DESCRIPTION

### A. Project Development Objective

#### PDO Statement

21. The Project Development Objective of the first phase (“the Project”) is “to increase access to resilient infrastructure and infrastructure sector jobs, and improve the capacity of select institutions.”

#### PDO Level Indicators

- Direct users that benefit from improved access to sustainable transport infrastructure and services (Number, disaggregated by gender and youth)
- People benefiting from climate resilient infrastructure (Number, disaggregated by gender and youth)
- Direct person-days of employment generated by construction activities (Person-Days, disaggregated by gender and youth)
- Government staff receiving training in core technical disciplines and receiving required licenses (as applicable) (Number)

### B. Project Components

22. Phase 1 consists of three components aligned with the MPA Pillars, as well as a fourth supporting implementation. Together, these components are designed to deliver early, visible results in high impact locations while embedding resilience, preparedness, and institutional strengthening as foundations for subsequent phases.

#### Component 1: Resilient Infrastructure for Connectivity, Jobs, and Growth (US\$138.2 million)

23. **Sub-Component 1.A: Safe and Resilient Air Access (US\$19.5 million)** - At international and regional airport hubs (Mogadishu, Baidoa, Kismayo and Garowe) investments will include (i) airfield ground lighting; (ii) passenger and cargo screening equipment; (iii) improved power supply and water supply; (vi) passenger security and baggage screening equipment to enable compliance with civil aviation regulations; and (vii) improved Airfield Rescue and Firefighting (ARFF) capacity to allow operations to meet international standards. Investments in domestic airports (Galkayo, Dhusamareb, Jowhar, Dhobley)<sup>11</sup> will improve the resilience of airfield infrastructure to extreme weather conditions and improve continuity of commercial and humanitarian operations for emergency response during climate-related disruptions, including (i) upgrade or rehabilitation of airstrips, (ii) provision of airfield drainage or swales to redirect water and protect airfield assets, (iii) installation or rehabilitation of fencing for perimeter security, (iv) development of small terminal or passenger facilitation services with improved power supply and water supply, and (vi) passenger security and baggage screening equipment.

24. **Sub-Component 1.B: Resilient Urban Infrastructure (US\$68 million)** – The subcomponent finances construction of approximately 2.87 kilometers of priority main trunk drainage line for Catchment Outfall O16 in Mogadishu to address chronic and intensifying urban flooding. Implementation of Catchment O16 is prioritized under Phase 1 due to its high impact on reducing recurrent flood risk and related disruptions.

25. **Sub-Component 1.C: Economic Corridors (US\$50.7 million)** – This subcomponent will finance a key missing link in one of the Horn of Africa Initiative corridors, connecting Hargeisa to the Ethiopian border at Ina Guha town.

<sup>11</sup> Airstrips selected based on a multicriteria analysis incorporating technical, economic, financial and other factors. The analysis prioritized nine total airstrips, with the three included in this phase chosen based on accessibility and readiness for investment. Additional airports will be included in future phases of the MPA.



Investments include a 63 kilometers section of roadway and associated facilities, provides a critical link between Hargeisa city, through Salahley district center, to the Ethiopian border.

## Component 2: Institutional Transformation and Human Capacity Building (US\$5.8 million)

26. **Sub-Component 2.A – Air Transport Capacity Development (US\$2.5 million).** This component supports the institutional, policy and technical capacities of the relevant institutions, both in the short term and setting a foundation for deeper reforms to be carried out throughout the life of the MPA. Activities will include:

- (i) **Somalia Airports Authority.** Preparation of a roadmap for establishing a Somalia Airports Authority, setting out: (a) the proposed legal mandate and governance/accountability arrangements; (b) roles and interfaces with key sector actors; (c) core operational and staffing requirements and capacity-building needs; (d) a phased, time-bound implementation plan; and (e) capacity building for current airport operations.
- (ii) **Ministry of Transport and Civil Aviation Technical Assistance.** Technical assistance supporting the MoTCA in strengthening its capacity for oversight of the sector, including (a) preparation of a MoTCA five-year strategic plan, (b) design of digital systems architecture to support drivers licensing and vehicle registration, and (c) preparation of a civil aviation masterplan. It will also finance targeted training to increase the number of fully licensed air traffic controllers, including women.

27. **Sub-Component 2.B – Construction and Engineering Capacity Building (US\$1 million)** – This subcomponent will address short-term capacity constraints while launching deeper reforms to be supported throughout the life of the MPA. Activities will include:

- (i) **Somali engineers' board.** Preparation of a roadmap for establishing an engineers' board, including (a) a baseline needs assessment, (b) benchmarking and peer learning from other similar bodies in the East African Community, (c) development of legal and regulatory framework documents, and (d) institutional design and operating procedures (including for financial sustainability).
- (ii) **Ministry of Public Works, Reconstruction and Housing Technical Assistance.** This sub-component will also support key institutional capacity building, such as (a) preparation of an axle-load law and (b) strengthening the ministry's Management Information System (MIS), and (c) preparation of detailed design and E&S assessments for a new ministry building (to be financed under a future phase of the MPA).

28. **Sub-Component 2.C – Preparation of Subsequent Phases (US\$2.3 million)** Led by the MoF, this subcomponent will support the planning of future MPA phases clarifying sequencing and design readiness, sharpening spatial and sectoral prioritization through transparent criteria, and strengthening investment frameworks. Activities will include:

- (i) **Basin of integration infrastructure needs assessment.** Analytical studies at the basin level, leveraging existing national data and platforms to develop a tailored assessment, including: (a) validation of the basin extents, (b) mapping of key supply chains and diagnosis of physical and systematic bottlenecks, (c) assessment of opportunities to drive sustainable job creation through infrastructure and service development, and (d) development of a regional infrastructure implementation plan. As part of this effort, the subcomponent will explore the deployment of the World Bank's Smart Parcel Atlas for Resilient Cities (SPARC), to strengthen risk informed spatial analysis, infrastructure prioritization, and investment sequencing for subsequent phases.<sup>12</sup>
- (ii) **Integrating Disaster Risk Management.** This activity will assess sub-national Emergency Preparedness Response systems in Somalia to identify priority investments and strengthen linkages with national institutions. It will support refinement and operationalization of the country's Disaster Risk Financing strategy and improve key disaster risk management capacities across selected federal member states and

<sup>12</sup> SPARC is a geospatial, parcel-level intelligence service that uses high-resolution satellite imagery, Artificial Intelligence, and global datasets—combined with local government data—to support land use and infrastructure planning, risk-informed urban development, and investment targeting.



municipalities. It will also support investment planning (Public Investment, PPP, asset-based revenue generation) for ensuring strategic infrastructure development.

### **Component 3: Sustainable Operations and Maintenance (US\$8.5 million)**

**29. This component will address critical challenges in the ongoing operations of the selected infrastructure sectors,** including airspace and airport management, resilient infrastructure operations and maintenance budgeting, as well as in securing sustainable revenue streams for operations and reinvestment.

**30. *Sub-Component 3.A – Investments in air traffic management and sustainable overflight revenue (US\$6.5 million)*.** The subcomponent contributes to a safer, more efficient, and revenue-generating Flight Information Region. Installation of specialized equipment, software and data services will improve airspace management throughout the country by providing Mogadishu's Aden Adde International Airport (AAIA) and select regional/international airports with modern communication, navigation and surveillance systems,<sup>13</sup> improving the efficiency of both commercial and humanitarian flights.

**31. *Sub-Component 3.B – Infrastructure Operations and Maintenance (O&M) (US\$2 million)*.** This sub-component will support the development of improved O&M plans for project-financed drainage infrastructure, roads and airports. These plans will factor in the available MoPWRH FGS, FMS, selected cities budget, as well as financing needs for maintenance of existing infrastructure assets. For airports, this will include options to leverage aeronautical pricing to ensure maintenance of assets and continuation of airport services. For urban drainage, maintenance committees composed of community members would be set up for all project supported infrastructure investments, i.e. Mogadishu drainage committee, and these committees would be responsible for informing the maintenance needs and mobilizing community members for maintenance work as needed. These committees will be trained under the Project by the MoPWRH and engineering supervision consultant.

### **Component 4: Infrastructure Implementation Coordination and Management (US\$7.5 million)**

**32. This subcomponent will establish project coordination and implementation mechanisms to facilitate coordinated and efficient infrastructure implementation for all phases of the MPA.** It is expected to establish a high level (and limited) coordination mechanism within the MoF to ensure alignment of all phases, as well as support for dedicated Project Implementation Units (PIU) within MoTCA and MoPWRH. It will leverage existing implementation structures within the MoPWRH and cities). Finally, noting that this will be the first time the MoTCA has served as an implementing entity of a World Bank project, it will support capacity building for project implementation.

**33. In addition, this component will support institutional capacity development for project and contract management within the implementing Ministry's civil service.** Phase 1 will support a technical and fiduciary skills diagnostic of civil service positions within relevant ministries to prepare a human resource needs assessment, review HR policies on job qualifications, develop staff learning programs, and outline a transition roadmap to enable internal government staff to implement future MPA phases. Subsequent phases will potentially support a gradual reduction of external consultants and an increase in regularized civil servants for project implementation.

### **C. Project Beneficiaries**

**34. The Project's direct beneficiaries will include;** (i) annual passengers of domestic and international flights in the airports/airstrips where the Project is investing in enhanced services and/or infrastructure (approximately 687,500 persons); (ii) those benefitting from flood risk reduction due to improved urban drainage (66,192 persons); (iii) the 280,000 users of the improved economic corridor, including passengers in public transport and private cars, as well as drivers of long-haul trucks providing international freight services. In total, the project is expected to directly benefit

<sup>13</sup> Automatic Dependent Surveillance – Broadcast/Very Small Aperture Terminal (ADS-B/VSAT); Automatic Weather Observatory Stations (AWOS), Aeronautical Information Management (AIM), and Air Navigation Service investments



917 people (of which 459,000 are women and 642,000 are youth). Beneficiaries will also include those benefiting from the Project's capacity building activities, including the MoTCA and Somalia Civil Aviation Authority (SCAA) staff, as well as the MoPWRH staff eligible for engineering certification. The Project's indirect beneficiaries will include passengers flying through Somali airspace benefiting from improved safety and security, as well as the 31,000 residents living within the catchment of the economic corridor.<sup>14</sup> Finally, considering the civil works required, the Project will benefit those working jobs in construction, direct jobs in operations and maintenance such as inspection, routine repairs, vegetation clearing, track maintenance, signaling or safety system check, runway and facility maintenance, equipment servicing, baggage handling, safety and security operations, and emergency preparedness.<sup>15</sup>

**35. In addition, the Project will have a direct and indirect impact on job creation.** Regarding direct jobs, it is estimated that the Project will create 600 construction sector jobs during project implementation. Operations and maintenance of these assets will create additional employment, such as an estimated 100 new airport operations staff, as well as those employed maintaining the economic corridor and trunk drainage. Finally, the Project is expected to indirectly generate jobs through induced economic development. Initial calculations estimate the Project will generate approximately 31,000 more and better jobs.<sup>16</sup>

#### **D. Rationale for Bank Involvement and Role of Partners**

**36. The World Bank has a well-established infrastructure portfolio in the Federal Republic of Somalia and extensive experience delivering operations in fragile, conflict-affected and violence-affected FCV context, playing a crucial role in convening stakeholders.** Lessons from the Bank's long-standing engagement in Somalia, as well as from comparable FCV settings globally, have been systematically incorporated into the design of the MPA, including the use of phased scaling, readiness-based selection of investments, adaptive implementation, and enhanced risk mitigation measures. The Bank has deep experience in transport and urban infrastructure and governance reforms across Africa and has demonstrated capacity to deliver results in highly insecure and access-constrained environments through flexible implementation arrangements, third-party monitoring, and digital supervision tools. It has also served as a central coordinator, bringing together government stakeholders (FGS, FMS, and city-level agencies) as well as United Nations (UN) agencies and private operators to align technical standards, manage risks, and ensure the continuity of service delivery.

**37. The MPA provides a platform to synchronize efforts with other development partners,** including the African Development Bank's US\$77 million in investments in Horn of Africa transport links, the United Kingdom, the European Investment Bank, the European Union, Denmark, and Qatar, aligning investments to mobilize finance for global public goods and creating an enabling environment for private capital through establishing concrete operations and maintenance structures and longer term, sustainable job creation. It also aligns with IFC and MIGA country programs to harmonize financing approaches and risk mitigation approaches in a challenging FCV context. With additional support through the Somalia Multi-Partner Fund (MPF),<sup>17</sup> the infrastructure portfolio reflects a phased scaled-up of World Bank engagement focusing on more transformative climate smart investments. This partnership approach has facilitated the careful assessment of needs, risks, and institutional capacity at different levels of government before scaling financing for investments.

<sup>14</sup> World Bank Corporate Scorecard methodology for "People that benefit from improved access to sustainable transport infrastructure and services"

<sup>15</sup> For future phases, beneficiaries will include both direct, and indirect jobs using accepted World Bank Methodologies including from the Corporate Scorecard Indicator Guidance note on More and Better Jobs for Infrastructure and the Somalia Jobs Report. The methodology for this Project, including the determination of baselines and multipliers will be developed during Phase 1.

<sup>16</sup> Note that estimates are based on the proposed methodology for the Jobs Corporate Scorecard Indicator, prepared on a pilot basis. For future phases, the indicator of 'more and better jobs' will be included with targets calculated using accepted World Bank Methodologies including from the Corporate Scorecard Indicator Guidance note on More and Better Jobs for Infrastructure and the Somalia Jobs Report.

<sup>17</sup> Includes multiple donors, including Foreign Commonwealth, and Development Office (FCDO), Kreditanstalt für Wiederaufbau (KfW), European Union (EU), Danish, among others.



## E. Lessons Learned

### 38. Key lessons leveraged in the design of the MPA, including Phase 1, emphasize:

- **Ensuring Somalia's FCV and political context are integrated in project design.** Somalia's operating environment creates unique risks for large-scale infrastructure delivery, including insecurity, disputed territorial control, an unfinished political settlement, fragmented governance, and low trust across institutions and levels of government. The MPA instrument reflects lessons from operating in this context.
- **Designing implementation arrangements that leverage synergies across all levels of government.** The Project adopts a decentralized model that empowers local authorities where feasible, while maintaining a central enabling role in policy, strategy, and regulatory oversight, drawing on experience from Somalia Urban Resilience Project Phase II (SURP-II, P170922). Clear implementation of responsibilities to mandated institutions helps build lasting capacity in the right places, strengthening legitimacy through visible results. To manage roles and ownership, agreements between federal, state, and local authorities will be discussed prior to implementation. The Standard Form of Agreement with UN agencies helps manage integrity and Environmental and Social Standards compliance risks through output-based deliverables, robust reporting, and stronger accountability, supporting supervision in access-constrained settings.
- **An iterative and phased approach will be critical to the Project's success in building institutional capacity.** Phase 1 will assess needs, risks, and capacity at FGS, FMS, and city levels before scaling investments, consistent with lessons from FCV and stabilization contexts on the importance of conflict- and access-aware planning. The approach supports use of country systems to strengthen accountability and state society relations, while enabling data driven decisions for future phases.
- **System-wide airport improvements can strengthen regional connectivity and broaden economic benefits.** Regional experience shows that network improvements reduce reliance on a single facility, improving continuity during weather, technical, or security disruptions.<sup>18</sup> They connect underserved areas to markets, reduce travel times where roads are constrained, and support jobs, local revenues, and investment in secondary cities, making air connectivity a practical resilience pathway for the Project.

## III. IMPLEMENTATION ARRANGEMENTS

### A. Institutional and Implementation Arrangements

39. **The Project will have three implementing agencies: the MoF, MoTCA, and MoPWRH.** Teams from these Ministries, jointly as a National MPA Coordination Unit, will be responsible for the overall implementation of all activities in the Project. For urban drainage activities, the Banadir Regional Administration (BRA) Project Implementation Unit established under SURP-II will be leveraged. The Project will support a gradual move away from stand-alone project structures and enable a transition to greater reliance on civil servants, supported by upfront transition planning during Phase 1 to strengthen long-term sustainability and capacity. Detailed implementation arrangements are outlined in Annex 1.

40. **Project governance will be led by two committees: a Ministerial Committee and a Technical Steering Committee.** The Ministerial Committee, chaired by the Minister of Finance and including the Ministers of the MoTCA and the MoPWRH, and the State Minister of the Office of the Prime Minister, will meet on a bi-annual basis to review the overall MPA program progress and set high level direction for ongoing and future phases. Day-to-day governance will be the responsibility of a Technical Steering Committee led by the Director General (DG) of the MoF and consisting

<sup>18</sup> For instance, West and Central Africa Air Transport Safety & Security Project (P100785) and Tanzania Transport Integration Project (P165660)



of the DGs of the MoTCA and the MoPWRH, as well as representatives from engaged FMS ministries/BRA depending on the agenda, which will meet on a quarterly basis, or more often as required.

## B. Results Monitoring and Evaluation Arrangements

**41. Resources allocated under Component 4 support project management activities, including Monitoring & Evaluation (M&E).** Project outcome and intermediate indicators will be measured using the methodologies outlined in the Results Framework. As the MoF will serve as overall coordinator for the Project, they will provide high level consolidation and reporting on M&E during the Project implementation cycle. Project information updates and data management will be coordinated across relevant departments. Contractors and supervising engineers will provide monthly site progress reports to validate work quality and Environmental, Social, Health and Safety compliance. Results and compliance assessments will provide feedback loops with government stakeholders and project committees, including community consultations and grievance redress at project sites. Semiannual project progress reports will be prepared and submitted to the World Bank no later than 45 days after the end of each calendar semester. The structure of these reports will be detailed in the Project Operations Manual (POM). In line with the experience across other World Bank-financed investments, the Project will benefit from enhanced support for capacity building for digital supervision and M&E, including through tools such as the Geo-Enabling initiative for Monitoring and Supervision that has been employed across the portfolio.

## C. Sustainability

**42. The air transport sector can generate substantial and sustaining revenue, but unlocking this potential and sustaining investments across other sectors will depend on stronger governance and reliable O&M funding.** Key equipment upgrades are expected to justify higher overflight fees, increasing revenue and budget flows to the MoTCA and SCAA and enabling commercialization and private sector participation, including establishing a Somalia Airport Authority. In parallel, the Project will support sectoral institutions to strengthen finance and management systems to fund recurrent O&M costs, retain qualified staff, and sustain service delivery. Doing so will build trust among stakeholders and motivate public sector decision makers to allocate sufficient O&M resources.

**43. Under Component 4, the Project will drive a gradual transition to government ownership and direct implementation of infrastructure projects.** Up to this point, nearly all infrastructure projects in Somalia (not only World Bank financed but supported by all development partners) have relied on project hired consultants for project management. As such, Phase 1 will launch a longer-term gradual transition to direct government implementation. Component 2 support for institution building and capacity development will also support this transition.

**44. The selection and design of all infrastructure activities in the Project were based on recently completed climate risk assessments.** These assessments, prepared by the World Bank as part of the recently completed CCDR, assessed the current and planned transport network in Somalia against climate hazards, creating a hierarchy of interventions necessary to ensure sustainable connectivity under adverse conditions. Critical risks identified include pluvial, fluvial, and coastal flooding. See the climate change section below for more information.

## IV. PROJECT APPRAISAL SUMMARY

### A. Technical, Economic and Financial Analysis

**45. Reinforcing the design of the MPA, Phase 1 activities have been selected based on three objectives:** (i) delivering early success to build stakeholder buy in, political momentum, and crowd-in other financing; (ii) setting a foundation for future investments through institutional development and hands-on capacity building; and (iii) facilitating reliable and safe access throughout the country to reduce implementation risk and costs for future phases. Therefore, activities have been selected to ensure rapid implementation, building upon existing projects with complementary



investments and/or design, hands-on implementation capacity building, and national level accessibility. All activities have been developed through a consultative process, including direct citizen engagement.

### Component 1.A

**46. The Project reinforces the Federal Republic of Somalia's compliance with mandatory international safety and security standards.** The subcomponent supports International Civil Aviation Organization (ICAO)-compliant interventions in critical infrastructure and equipment, emergency response, and access control. Project costs are informed by data provided by SCAA, United Nations Humanitarian Air Services (UNHAS) and the Simple Network Time Protocol (SNTP) Lab Analysis that was collected for an airport network level climate vulnerability analysis to determine exposure risk to estimate damage losses and reconstruction cost.

**47. A multicriteria analysis was developed as a decision-making tool to evaluate and prioritize project airports.** The framework was consulted and agreed between the FGS and FMS representatives. The criteria included: population catchment, food insecurity, aircraft movements, level of effort to meet planned airport code, number of days closed due to climate events, socio-economic impact, security risk.

**48. Airport Upgrades:** Interventions are being designed to meet minimum requirements for runway and airstrip length, stripe width, obstacle limitation surfaces, apron and taxiway dimensions, and perimeter security to meet to ICAO Category 3C aerodrome standards.<sup>19</sup> Based on an assessment of gaps to meet 3C standards, site specific improvements at each aerodrome have been identified, including:

- *Improved airfield runways and drainage (US\$9.3 million)* – Responding to the critical climate vulnerabilities of Somalia's airstrips, four airports (Dhusamareb, Jowhar, Dhobley, Galkayo) have been selected for runway and drainage improvements under Phase 1. Investments include full rehabilitation of the runways at two of these airports (Jowhar and Galkayo). Runway designs meet the geometric requirements for Code C compliance, i.e. runway width (30 meters), runway shoulders (7.5 meters each), runway strip width (150 meters), and runway end safety areas (240 meters), as well as all taxiway and apron upgrade dimensions. Building off of a national level climate vulnerability study which highlighted key risks at all four selected airports, drainage improvements at all airports will account for increasing vulnerability to flooding, with drainage designs based on site specific flood modeling. NBS for airfield drainage will complement hard investments in capacity.
- *Airfield ground lighting (US\$3 million)* – LED ground lighting has been prioritized for two airports (Mogadishu and Garowe) to allow diversion flights during airspace emergencies, improved visibility during extreme weather events, and adaptive capacity of the airport system. These improvements are critical as part of the overall initiative to improve the safety and resilience of the air transport sector.
- *Terminal and tower improvements (US\$2.95 million)* – To support improved passenger safety and security, terminals with towers will be constructed in Dhobley and Dhusamareb. Terminal designs accommodate 100 passengers, including waiting areas, furnishing, passenger screening equipment, solar power supply, and reliable water supply. In Jowhar, where a terminal is already under construction, the Project will support installation of key furnishings and screening equipment. In Garowe, Baidoa and Kismayo, the Project will support installation of improved power, water and security systems.
- *Perimeter fencing (US\$0.55 million)* – In order to prevent incursion of the airport space by wildlife and community members, presenting significant safety and security challenges, the existing fencing at all four local airports will be improved. This will also allow ICAO certification.

**49. Airfield rescue and firefighting (ARFF) (US\$3.7 million).** ARFF vehicles and/or equipment acquisitions will be in accordance with each airports ICAO ARFF category based on performance criteria (e.g., engine torque, top speed,

<sup>19</sup> ICAO Annex 14, Volume I – Aerodrome Design and Operations



acceleration rates to reach incident sites within defined response-times), pumping capacity (e.g., tank volumes, minimum output in liters per minute), chassis and driveline configurations, crew safety, and other criteria.

### Component 1.B

**50. Mogadishu resilient urban trunk drainage infrastructure.** The urban area trunk drainage extension (of approximately 2.87 kilometers) to the current drainage network will contribute to stormwater collection to reduce urban flooding. Enhancements also include expanding the capacity of runoff collectors/outfalls leading to the sea, additional conveyance lines,<sup>20</sup> increasing the number of inlet points for improved load handling and installing infiltration trenches, adding retention ponds to increase buffering during heavy rainfall. The interventions will pilot nature-based solutions for localized stormwater management where feasible, such as planting native grasses in the retention pond, to mitigate drought impacts by enhancing water retention and reducing the urban heat island effects. Beyond flood risk reduction, the subcomponent will generate employment through labor-intensive construction and longer-term operations and maintenance, stimulate local construction supply chains, and protect businesses, markets, and household assets, supporting safer, more reliable urban economic activity, particularly for internally displaced persons and the urban poor. The investments will thus reduce economic losses associated with urban flooding and at the same time enhance the resilience of rapidly growing cities to increasingly frequent extreme rainfall events. Specific investments will include: (i) upgrading and/or installation of flood mitigation infrastructure with the protection level of a 1-in-25-year event, including NBS where feasible; (ii) construction of flood control structures including dykes, polders, manholes, and drainage; and (iii) laying out the foundation for future phases on establishment of a real-time flood warning system. Flood mitigation benefits will be sustained through complementary operations and maintenance, including routine cleaning, desilting, and solid waste management.

**51. A network layout has been delineated which optimizes the efficiency of flow paths, minimizing need for excavation and trenchless (no-dig) technologies use.** The planned network layout and sizing will avoid the need for mechanical pumping for discharge at sea. The system has been designed using a design rainfall event with a return period of 25 years characterized by a daily precipitation of 134 millimeters and an intensity of 124 millimeters/hour. The trunk drain extends approximately 2.87 kilometers and is designed to convey a peak discharge of 82.54 meters<sup>3</sup>/second. The proposed design incorporates KM Pond 4 (24,000 meters<sup>3</sup>) as a temporary storage facility, which helps attenuate peak flows and reduce hydraulic stress on the downstream drainage network. In addition, the system enables integration with serviceable secondary drainage lines to be maintained by government, transforming the investment from a standalone conveyance structure into a fully functional drainage network. This integrated approach significantly enhances system performance by rapidly expanding the effective catchment area and improving overall flood management capacity. The functioning of the outlet system was verified for a few scenarios: (i) normal scenario: mean sea level as downstream boundary condition; (ii) high tide scenario: +1.0 meters as downstream boundary condition, and (iii) worst case high tide scenario: +2.5 meters as downstream boundary condition.

### Component 1.C

**52. The Haleya–Ina Guha Economic Corridor is a strategic transport link in Somaliland, linking Hargeisa with the Ethiopian border.** The corridor forms part of the Ethiopia–Somalia Regional Economic Corridor, part of the Mogadishu, Berbera and Bosaso corridor identified as part of the Horn of Africa Initiative. This corridor is strategically important as it provides a direct connection between Somalia and Ethiopia, facilitating cross-border trade, regional integration, and access to international markets. The corridor begins at Haleya town, approximately 16 kilometers east of Hargeisa, and extends southward to Ina Guha, a border town with Ethiopia, covering a total distance of about 91.5 kilometers. Serving both local access and international freight, the corridor carries significant heavy traffic, including an average of 416 heavy trucks per day, as well as several thousand light vehicles, with traffic demand projections estimated at

<sup>20</sup> Trunk drainage will exclude secondary lines which together with infiltration trenches will likely be supported through subsequent financing.



4.3 percent growth per year. The existing road condition is poor to extremely poor, with a combination of deteriorated asphalt and gravel sections, frequent potholes, erosion, and inadequate drainage. Floodplains and seasonal water crossings regularly disrupt traffic, making the road unreliable and unsafe. A climate change assessment identified that the fluvial and pluvial flood risk faced by the corridor will continue to increase.

**53. The section selected for financing supports the core traffic of the corridor: a 63 kilometers section from Isha Gobka, just south of Hargeisa city, to the Ethiopian border.** This leverages ongoing work is already under implementation by the Hargeisa Municipality, including a 6.46 kilometers section linking the southern side of Hargeisa to Isha Gobka. Selection of this section for phase 1 followed extensive consultations with local and regional stakeholders, supporting trade traffic from the Berbera port bound for Ethiopia, as well as establishing a resilient link between Hargeisa, Salahley District, and other communities along the corridor. Implementation will leverage an arrangement with a UN Agency which is already implementing road construction activities in the region, financed by another World Bank project.

**54. As Somalia does not yet have its own road design standard, the primary design standard adopted is the Ethiopian Roads Authority (ERA) Manuals, which govern the geometric design, pavement design, drainage, and road safety requirements.** The design places a high priority on durability and cost-effectiveness through the utilization of locally sourced materials, strictly adhering to the ERA Design Class DC-6 standards. Compliance with ERA standards is also important to guarantee the road continuity to Ethiopia as it crosses the border. The road design includes a 7 meters wide asphalt concrete carriage way (two-lane, two-way traffic) with 2 meters wide shoulders on each side for safety. In settlements and junctions, the design accommodates paved shoulders, pedestrian walkways, and solar street lighting to support the needs of all users. Side drains have been designed to accommodate the site-specific demand, accounting for future climate-driven flood risk.

**55. A nationwide climate risk assessment identified this corridor as potentially vulnerable to increased fluvial and pluvial flooding.** As a result, a more detailed climate vulnerability assessment was conducted as part of the corridor design process, identifying key locations mitigation measures. Key improvements to the design to ensure climate resilience include: (i) enhanced drainage (including increased capacity, elevated embankments, and scour protection works), (ii) elevated roadways in flood prone areas, (iii) stormwater catchment in settled areas to support community resilience, (iv) use of thermally stable pavement materials to reduce extreme heat risk, and (v) leveraging NBS to reduce runoff and mitigate extreme rainfall. These measures increase the cost of the investment by US\$8.88 million, although their inclusion will reduce maintenance costs by up to 40 percent over its 20-years design life.

## Component 2

**56. Ministry level support will help development of implementing agencies.** Within the MoPWRH, the Project will carry ongoing capacity building activities in World Bank-financed SHIP and SURP II projects forward, such as the development of an Axle Load law and rollout of a ministry wide Management Information System to allow for the digitalization of several permitting and operational processes. It will also support the physical capacity of the ministry, including preparing the designs and E&S instruments for a new ministry building. In the MoTCA, the Project will contribute to already established capacity building and policy development plans. Training, upskilling, and licensing in air navigation services and air traffic control will help raise the share of SCAA professionals holding required licenses, supporting international compliance. It will also finance feasibility and design studies for air transport investments in future MPA phases. In the MoF, identified activities include support for operationalization of a Disaster Risk Financing Strategy, as well as assessments and technical assistance for revenue generation and cost recovery of infrastructure assets, operations and maintenance financing frameworks, and Public Investment Management.

**57. Support for the development of new institutions will include the initial design of a Somali engineers' board and a Somali Airports Authority.** While the institutional development needs in Somalia extend far beyond these two, they have been selected for support under Phase one due to four considerations: (i) they have been identified as major



bottlenecks for sectoral development; (ii) they are expected to generate additional revenue, reducing or eliminating any potential burden on the national budget; (iii) they enjoy strong buy-in from relevant stakeholders; and (iv) they are expected to support increased private sector involvement in the infrastructure sector. The Somali engineers' Board is critical for professionalizing the engineering profession, setting clear and verifiable criteria for the practice of engineering, and unlocking qualified Somali engineers' ability to work within the country and region. The Somali Airports Authority will serve a critical role in sustainable development, operations, and maintenance of key airports in the country, providing an important governance role in establishing concessions with private service providers and as a vehicle to ensure revenue generated through landing and other fees is leveraged for maintenance and further development. It will also address a critical issue identified by ICAO, as SCAA currently serves as both the operator and regulator of airports, creating serious conflicts of interest.

**58. In addition, this component will support the preparation of future phases of the MPA.** It will support a targeted assessment of the country's key economic basins, to identify key economic drivers and the potential bottlenecks to their growth and impact on job creation. These assessments will inform the selection and design of investments under future phases of the MPA.

### Component 3

**59. Air Traffic Management.** The Communications, Navigation, Surveillance (CNS) system standards<sup>21</sup> set the foundations for safe, reliable, and continuous operations across the Somalia's airspace. The Project will supply, install, integrate, commission, and provide technical support for Automatic Dependent Surveillance – Broadcast/Very Small Aperture Terminal (ADS-B/VSAT). ADS-B (1090 Megahertz Mode S Extended Squitter) ground surveillance system will enhance air traffic control and improve visibility of commercial and private aircraft operating in Somali airspace. The VSAT installations and acceptance testing will comply with the relevant international standards and guidelines and protect all surveillance data from interception or unauthorized access. Each remote VSAT terminal will connect to the SCAA headquarters via satellite. Cybersecurity measures include role-based access control for system users, integration with the SCAA's security monitoring system, and encrypted administration. Small scale solar systems (50 kilowatt) to be deployed at all airport sites installing ADS-B/VSAT equipment, which will require Uninterrupted Power Supply (UPS) backup for 30–60 minutes, as well as an auto-start generator. Clear acceptance testing protocols, documentation, and warranties will also be included in the technical requirements. Preventive Maintenance will ensure twice-yearly site inspections and annual calibration, as well as service-level agreement for remote support in 4–8 hours, on-site support within 48–72 hours, depending on location. Investments may also include other equipment as required.

**60. Operations and Maintenance.** Identified by various previous studies as a critical next step in the country's infrastructure development, establishing dedicated and reliable structures for infrastructure operations and asset maintenance will be critical. This is especially through as the MPA promises to significantly contribute to Somalia's physical capital. For airports, this will include options to leverage aeronautical pricing to ensure maintenance of assets and continuation of airport services. For urban drainage, maintenance committees composed of community members would be set up for all project supported infrastructure investments.

### Climate Change

**61. Climate Risk Screening.** World Bank climate-risk screening rates the Project's current hazard exposure as substantial but anticipates it will fall to moderate after implementation. Somalia's climate risk profile for transport and urban systems is characterized by intensifying floods, droughts and extreme heat with emerging coastal hazards. These are amplified by rapid urbanization and FCV dynamics. Mogadishu, and secondary cities such as Hargeisa, Kismayo, Baidoa, Beledweyne, Dhusamareb, and Garowe, face high flood risk driven by intense rainfall,

<sup>21</sup> ICAO Annex 10 Aeronautical Telecommunications (Vol. I–V)



inadequate drainage, and encroachment into floodplains and hazard-prone urban areas. Airports confront material risks from extreme rainfall and pluvial flooding (runway and apron inundation), storm surge and sea-level rise at coastal sites, and extreme heat that degrades pavements and equipment reliability. A transport risk assessment prepared for the Somalia CCDR shows pluvial flooding as the most damaging hazard, impacting 20 of 21 assessed airports.<sup>22</sup> The multi-criteria analysis for airport selection includes climate vulnerability criteria.

**62. The Project aligns with the Federal Republic of Somalia's climate commitments and the goals of the Paris Agreement, supporting both mitigation and adaptation objectives.** The proposed MPA and Phase 1 project contribute to Third Generation Nationally Determined Contribution (NDC3.0) and other national climate strategies. For mitigation, activities under 1.A (air transport) are classified as low risk since the component invests in improving airfield runway and drainage and safety and security at select regional and local aerodromes to enable network resilience. Activities under 1.B (urban trunk drainage) and 1.C (road rehabilitation and upgrading without capacity expansion, e.g., two-lane, two-way traffic) are Universally Aligned.

**63. For adaptation, airports and road sections financed by the MPA have been selected based on multi-criteria analysis, including climate resilience criteria, recognizing exposure to pluvial flooding and increasingly vulnerable to fluvial flooding.** Design features strengthen the resilience of critical transport infrastructure to flooding and extreme weather events while helping maintain connectivity for passengers, trade, and humanitarian logistics during climate shocks. Mogadishu is prone to acute and chronic flooding due to intense seasonal rainfall, limited stormwater drainage capacity, flat topography, and rapid urban expansion into low-lying areas. Short-duration heavy rainfall events frequently overwhelm existing drainage systems, leading to flash flooding, while inadequate drainage and poor runoff management result in persistent water accumulation in parts of the city. For airfield upgrade (1.A), asphalt concrete paving will be applied, airfield drainage infrastructure capacity improved, and NBS (e.g. swales to redirect water and protect airfield assets) applied. Airport meteorological stations will contribute to advance detection and historical data to monitor and manage extreme weather events. Trunk drainage enhancement (1.B) is designed in line with the Mogadishu Drainage Master Plan and engineered to manage a 25-year storm event, including the expansion of the capacity of runoff collectors/outfalls, extension of trunk drainage to improve stormwater collection, and pilot NBS for localized stormwater management. The road corridor section (1.C) incorporates US\$8.88 million in climate adaptation measures beyond the standard road design, as outlined above.

**64. For mitigation, modernized air navigation systems enable more precise and optimized flight routes, allowing aircraft to fly shorter, more direct paths that reduce fuel consumption and associated Carbon Dioxide (CO<sub>2</sub>) emissions.** Improved upper airspace management can guide aircraft to avoid regions of the atmosphere where contrail formation is most likely, reducing the non-CO<sub>2</sub> climate warming effects of aviation. By enabling better airspace efficiency through reduced holding patterns, ground delays, and inefficient routing, upgraded air navigation infrastructure can collectively deliver reductions in the aviation sector's overall carbon footprint at scale.

## Gender

**65. Women and girls are disproportionately affected by unemployment and weak economic integration.** Women, youth, persons with disabilities, and other vulnerable groups in Somali cities face disproportionate impacts from limited access to climate-resilient infrastructure and services, which constrains mobility, access to services, safety, and economic opportunities. Recurrent flooding and the resulting poor connectivity disrupt daily livelihoods, heighten risks during extreme weather and displacement, and reinforce structural barriers. In Somalia, just over half of the working-age population participate in the labor market, with low rates for women (43 percent) and youth (39 percent).<sup>23</sup> Women represent only 3.62 percent of total employment in the transportation and storage sector,

<sup>22</sup> World Bank assessment

<sup>23</sup> *Somalia Country Economic Memorandum: Towards and Inclusive Jobs Agenda*. World Bank. June 2021



reflecting deep structural gender gaps in access to technical and operational roles.<sup>24</sup> Within SCAA, women account for 19.76 percent of employees; but participation is lower in Air Traffic Management (ATM), where women represent only 13 percent of the workforce. Among these, around 55 percent hold a zero rating (compared with 41 percent of men), meaning they are not operationally deployable, and only a very small proportion are fully licensed to handle complex or multi-sector operations. Women remain underrepresented in transport due entrenched gender stereotypes that discourage participation in Science, Technology, Engineering and Mathematics (STEM) fields and reinforce employer biases regarding "masculine" roles. During the school-to-work transition, women also face information asymmetries, limited access to the professional networks, and insufficient quality training. In highly technical functions such as ATMs, these barriers result not only in lower entry rates but also slower progression toward full operational certification (from "zero rating" to fully licensed status). At the same time, there is an opportunity to equip women with specialized skills in emerging sectors, such as aviation, where demand for technical expertise is increasing.

**66. The Project addresses these constraints through several channels:** a skills demand assessment in the aviation sector, including a labor profile across SCAA, airport management, and Air Navigation Service providers, as well as a career cycle analysis of barriers to women's participation, to inform the design of training programs at different skill levels for subsequent MPA phases; and proactive outreach by the engineers' board/MoPWRH and Memorandums of Understanding with schools and universities to raise awareness and promote female participation in engineering careers. The Project will also place a specific focus on ATM capacity building to increase the number of women who are fully licensed and operationally deployable. Progression from zero rating to higher certification levels (up to three ratings) leading to full licensing significantly improves employability, operational responsibility, and career progression. Through the Project an additional 10 women will become fully licensed, increasing the share of women among all fully licensed staff from 7 percent to 30 percent. Licensing will be delivered through an ICAO-recognized institution in East Africa, with comprehensive support provided to women, including training costs, travel, and stipends. Gender sensitization will also be conducted within SCAA, including for supervisors of participating women. Licensing enables women to handle more complex operations, supervise multiple sectors, and access higher-paying positions, with potential salary increases above 100 percent from zero rating to full certification.

## Economic Analysis

**67. Airport Improvements:** The airport analysis quantifies the direct costs in infrastructure, safety and security, monitoring, O&M, and training. Costing is informed by sites specific assessments and benchmarking of unit costs with similar projects in Somalia. Operational data on aircraft movements, passengers, and landing charges are forecast over a 23-year horizon with a 3.6 percent annual growth rate. The quantified benefits comprise fleet operation cost savings, travel time savings, and Greenhouse Gas (GHG) emission reductions, with remaining benefits such as improved cargo handling, improved emergency and humanitarian response, and socio-economic spillovers (jobs, market access) acknowledged as additional. The cost-benefit analysis (CBA) over 23 years (a 3-year construction period) with a 6 percent discount rate shows the Net Present Value (NPV) of US\$16.3 million and an Economic Internal Rate of Return (EIRR) of 14.5 percent, with a Benefit-Cost Ratio (BCR) of 1.75. A sensitivity analysis with a 20 percent cost overrun and a 20 percent reduction in benefits demonstrates that the results remain highly positive (EIRR=8.2 percent). The switching value analysis indicates that NPV would reach zero only if project costs were to increase by 177 percent above the base case. The inclusion of shadow carbon pricing under both low and high price scenarios results in NPVs of US\$25.0 million and US\$33.7 million and EIRRs of 17.9 percent and 21.0 percent, respectively.

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<sup>24</sup> ILOSTAT (2022)



68. **Urban drainage.** CBA has been conducted for the proposed O-16 trunk drainage serving the L2 catchment in Mogadishu (2.87 kilometers, 9,456 households, 66,192 persons using a 250m buffer<sup>25</sup>) to assess the value of anticipated flood risk reduction benefits relative to the estimated US\$68 million investment cost. The analysis quantifies four benefit streams: avoided property damage to residential buildings (US\$5.15 million per year based on a 45 percent flood reduction), avoided household income loss (US\$98,000 per year), construction-phase employment (US\$247,000 one-time), and avoided damage to public infrastructure including roads, the airport, WASH and health facilities, and schools (US\$421,000 per year). At a 6 percent discount rate, the investment yields a positive NPV of US\$37.7 million, an EIRR of 9.4 percent, and a BCR of 1.53. The analysis applies an Expected Annual Damage (EAD) framework, integrating flood exposure modeling, and a US\$193 million ground floor asset base derived from the hex construction cost grid developed as part of an ongoing World Bank technical assistance. Benefits grow at 4.3 percent per annum (population plus urbanization). The quantified benefit streams do not capture several important channels which will improve the quality of life for beneficiaries, including avoided displacement costs, reduced waterborne disease incidence, lives saved, household contents losses, and urban land value effects.

69. **Economic Corridor.** CBA was undertaken for the Economic Corridor investments proposed under Component 1.C, applying the HDM-4 Road User Cost model to estimate reductions in vehicle operating costs, travel time savings, improvements in road safety, and changes in GHG emissions. Over a 20-year evaluation horizon and using a 6 percent discount rate, the Project yields a NPV of US\$38.6 million (of which the estimated road safety benefits are US\$8.2 million) and an EIRR of 15.2 percent. Sensitivity testing, incorporating a 20 percent increase in costs and a 20 percent reduction in benefits, indicates that the results remain economically robust, with a NPV of US\$24.5 million an EIRR of 11.1 percent. The inclusion of shadow carbon pricing under both low and high price scenarios results in NPVs of US\$37.4 million and US\$36.1 million and EIRRs of 15.0 percent and 14.7 percent, respectively. The Road Safety Screening and Appraisal Tool was used to identify and evaluate the potential traffic and road safety risks arising from the Project activities, with the Project interventions expected to reduce crash fatalities by 21 percent.

70. **GHG accounting.** Somalia's total GHG emissions are low, estimated at 41 MtCO<sub>2</sub>eq in 2020, representing less than 0.03 percent of total global emissions, although per capita emission are not negligible (at around a third of the global average) and have the potential to grow rapidly.<sup>26</sup> The Project is assessed as contributing a net reduction in GHG emissions of 197,833 MtCO<sub>2</sub>eq over the life of the investments, or 8,403 tons per year. Key contributions to the Project's GHG impact include the aviation investments which will enable the use of more efficient aircraft, modern avionics, improved routing, and upgraded navigation systems, as well as the economic corridor. The Project does not prevent a future transition to lower carbon alternatives.

71. **Financial Analysis.** The aviation sector accounted for 9.4 percent of all domestic revenue in 2024. In that year, revenues totaled US\$34.8 million, with the majority coming from overflight fees, supplemented by airport fees, ticket sales tax, and passenger charges. The government also benefits from other airport fees and ticket sales taxes, including US\$3.9 million in airport fees and US\$6.4 million in ticket sales tax (about 14 percent of the overall sales tax for 2024). The Civil Aviation Division charges US\$15 per departing international passenger, generating US\$2.2 million in 2024. Initial estimates have indicated substantial opportunities to increase generated revenue by aligning with international best practice, including standardizing the charging methodology<sup>27</sup> for overflight fees, which could triple revenues.

<sup>25</sup> The 250 m buffer was selected as an intermediate, pragmatic estimate of expected beneficiaries. It balances topography-driven drainage effects with less spatially deterministic factors such as road accessibility and indirect benefits from reduced flooding along mobility corridors. Future investments in secondary drainage will significantly magnify this impact (up to 23,000 households, 140,000 persons).

<sup>26</sup> The Federal Republic of Somalia (2022). Somalia's First Biennial Update Report to the United Nations Framework Convention on Climate Change (UNFCCC). Ministry of Environment and Climate Change (MoECC), Mogadishu, Somalia.

<sup>27</sup> Considering aircraft weight, distance travelled, and cost recovery.



## B. Fiduciary

**72. Financial Management Arrangements.** Project funds will be channeled through the Federal MoF to a designated account (DA) at the Central Bank of Somalia. The Project will utilize Government public financial management systems for budgeting, accounting, funds flow/banking, financial reporting, and audit oversight, with support from technical assistance that includes explicit knowledge transfer requirements. FM arrangements will leverage the FGS's centralized fiduciary platform by working closely with the External Assistance Fiduciary Directorate (EAFD) housed in the Offices of the Accountant General. Quarterly interim financial reports (IFRs) will be generated directly from the core Public Financial Management (PFM) system and submitted within 45 days of quarter end, and annual external audits will be carried out by the Office of the Auditor General in accordance with the PFM Act and Cash Basis International Public Sector Accounting Standards. This project will adopt the FundsChain platform to enhance transparency, traceability, and efficiency in the Project financial management. FundsChain will be integrated with Somalia Financial Information Management System (SFMIS), using payment flows alongside FundsChain's reporting capabilities. In addition, the platform will be used to track built and procured fixed assets, enhancing the expenditure reporting and end-to-end traceability of project resources within the three implementing entities. The Project will follow the transaction-based method of disbursement, which will include advances, reimbursements, direct payments, and special commitments. An initial advance will be disbursed to the DA following submission of withdrawal application to the World Bank accompanied by a six-month cash forecast. Subsequently, the Project will draw down on the proceeds of the grant following submission of an application for withdrawal accompanied by statement of expenditure (SOE) incurred. The implementing agencies will manage these processes in accordance with the World Bank's procedures. Withdrawal applications will be submitted via Client Connection, and disbursements will be made based on verified eligible expenditures. One DA in US\$ managed by the MoF will be opened at the Central Bank of Somalia in the name of the Project. All deposits and payments from this account will adhere to the provisions outlined in the Financing Agreement and Disbursement and Financial Information Letter. Authorized signatories will be designated to ensure compliance with World Bank disbursement procedures, and a grievance redress mechanism will be established to support transparent fund flows.

**73. The FM risk is assessed as High.** The fiduciary environment in Somalia remains challenging, with some potential levels of mismanagement, fraud, lack of transparency, and corruption. Although anticorruption and public sector regulations are in place, problems persist, contributing to low levels of trust in government institutions. Appropriate risk mitigation measures, including periodic extended implementation support activities complemented by close monitoring, will be undertaken in addition to providing on-the-ground fiduciary capacity support to the Project. The Project will ensure that the Project teams always have in place dedicated procurement and Financial Management specialists who are competitively selected and adequately trained jointly with the EAFD Unit and other relevant staff within the Office of the Accountant General. A series of on-the-job fiduciary training and the World Bank's periodic review and implementation support will be part of mitigation measures.

**74. Procurement Arrangements.** Procurement will follow the World Bank's *Procurement Regulations for IPF Borrowers: Procurement in Investment Project Financing - Goods, Works, Non-Consulting and Consulting Services* (7th Edition, September 2025) and be subject to the Bank's Anticorruption Guidelines dated October 15, 2006, revised in January 2011, and as of July 1, 2016. Procurement plans will be registered in the Systematic Tracking of Exchanges in Procurement (STEP). Guided by the Federal Republic of Somalia's country specific thresholds, methods will include open international and national Request for Bids for civil works, using the Bank Standard Procurement Documents for contracts subject to international competition; Request for Quotations for small works and goods; and competitive selection of consulting firms and individual consultants using standard Bank methods, with prior- and post-review to be specified in the Procurement Plan in STEP. For internationally competitive procurement packages estimated to cost more than US\$10 million, an early market engagement shall be carried out, with prior review by the Bank. Direct



selections for specialized/non-consulting interventions will be minimized but may include the use of the World Bank Standard Form of Agreement for UN Agencies for output-based delivery.

**75. The Procurement risk is assessed as High.** The main risks identified in the procurement capacity assessment are (i) the Project's broader scope and multiple ministries may complicate coordination and turnaround times; (ii) variations in institutional capacity across the MoF, MoPWRH, and MoTCA may lead to inconsistencies in procurement documentation, evaluation quality, and contract administration, and (iii) inadequate capacity for contract management may result in lack of adherence to contractual obligations, poor quality, cost and time overruns. To ensure that procurement is carried out satisfactorily, mitigation measures to bolster implementation capacity will be applied, such as: (i) maintain centralized fiduciary oversight within the MoF and conduct periodic inter-agency coordination reviews to monitor procurement progress; (ii) Hands-on Expanded Implementation Support (HEIS), to be further discussed with the client; (iii) mobilizing qualified consultants for procurement handling (and contract management), including engaging specialized technical backstopping support, where necessary, from institutions such as ICAO, the World Food Program, and the United Nations Office for Project Services (UNOPS) to strengthen preparation and review of aviation-related designs and technical specifications; (iv) the inclusion in the procurement section of the Project Operations Manual formalizing the coordination mechanism between the Project teams, detailing the roles, responsibilities, and time frames for carrying out the procurement functions; and (v) training.

**76. Market Situation:** The operational environment is characterized by significant infrastructure shortages, susceptibility to climate-related issues, and a high implementation risk. Security concerns, logistical obstacles, and restrictions on seasonal access may hinder the deployment and oversight of contractors. Potential impact of these on procurement processes include (i) lack of certainty that foreign companies are willing to participate in the procurement processes, given the security context of the country; (ii) the capacity and experience of domestic suppliers/contractors can be limited; (iii) Socio-economic and environmental aspects will be taken into consideration to align with sustainability principles; (iv) higher bid prices due to security-related costs, insurance premiums, logistics challenges, and risk contingencies incorporated by bidders; and (v) logistical and supply chain constraints, including import dependency for specialized equipment and construction materials, which may affect delivery timelines and pricing. These market related constraints necessitate early market engagement to understand the size, composition, and capabilities of the target market, and the identification of factors that could contribute to successful procurement.

### C. Legal Operational Policies

| Legal Operational Policies                  | Triggered? |
|---------------------------------------------|------------|
| Projects on International Waterways OP 7.50 | No         |
| Projects in Disputed Area OP 7.60           | No         |

### D. Environmental and Social (E&S)

**77. The Program has an overall E&S Risk Classification of High,** reflecting the nature, scale, geographic footprint, and implementation context of Phase 1 investments which will finance large-scale civil works for urban drainage, economic corridor road upgrading, and aviation infrastructure rehabilitation in a fragile and conflict-affected environment. Impacts on biodiversity and ecosystem services may arise from land clearance, habitat disturbance, and changes in drainage or land use, particularly along corridor sections traversing peri-urban and rural areas. Cultural heritage risks



are expected to be limited but cannot be ruled out, requiring screening and application of chance-find procedures during construction.

**78. Environmental and social risks.** Environmental risks are related to construction-induced impacts, including air and noise pollution, dust generation, solid and hazardous waste, wastewater discharge, and increased demand for water, energy, and construction materials. The large footprint of corridor upgrading raises the potential for cumulative and indirect impacts on ecosystems, hydrology, and surrounding communities, while specific risks for the aviation facilities include health and safety risks at the airports' fuel storage facility; life and fire safety of airport users as public access buildings; emergency preparedness and response and potential legacy soil and groundwater contamination. Environmental risks and impacts are further compounded by weak regulatory oversight, limited hazardous-waste management capacity, and climate-related stressors, including flooding and extreme rainfall, which can exacerbate impacts if not adequately managed. On social side, the Project faces high risks associated with community health and safety, land acquisition and resettlement, labor influx, and social cohesion. Civil works may lead to temporary or permanent physical and economic displacement, disruption of access and livelihoods, and increased exposure of communities to traffic hazards, accidents, and pollution. Somalia's complex land tenure systems, prevalence of informal land use, and presence of vulnerable groups including Internally Displaced Persons (IDPs), minority clans, women, and youth heighten the risk of exclusion, grievances, and conflict if impacts are not carefully managed.

**79. The Project also presents elevated risks of gender-based violence (GBV),** sexual exploitation and abuse, and sexual harassment (SEA/SH), driven by labor-intensive works, labor influx, power imbalances, and limited access to survivor support services. Security-related risks, including the potential use of security personnel to safeguard project assets, further increase the complexity of managing community safety in certain locations.

**80. Institutional capacity constraints, including uneven experience across implementing agencies and limited national and subnational E&S enforcement systems, represent a cross-cutting risk that could affect timely and effective mitigation.** Pillar 3 introduces additional risks related to operational sustainability, requiring robust operations and maintenance arrangements, revenue mechanisms, and institutional capacity to ensure compliance with E&S requirements over the life of the assets.

**81. These risks will be managed through a comprehensive ESF-based approach,** including site-specific Environmental and Social Impact Assessment (ESIAs)/Environmental and Social Management Plan (ESMPs), Resettlement Action Plan (RAPs), Labor Management Procedures, Stakeholder Engagement Plans, Security Management measures, and SEA/SH mitigation actions, supported by enhanced supervision, third-party support, and capacity building. The above assessments and plans for all required activities are being finalized, with disclosure required prior to the commencement of civil works. Continuous stakeholder engagement and accessible grievance redress mechanisms, including GBV-specific channels, will be critical to maintaining community trust and managing risks throughout implementation.

## V. GRIEVANCE REDRESS SERVICES

**82. Grievance Redress.** Communities and individuals who believe that they are adversely affected by a project supported by the World Bank may submit complaints to existing project-level grievance mechanisms or the Bank's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project affected communities and individuals may submit their complaint to the Bank's independent Accountability Mechanism (AM). The AM houses the Inspection Panel, which determines whether harm occurred, or could occur, as a result of Bank non-compliance with its policies and procedures, and the Dispute Resolution Service, which provides communities and borrowers with the opportunity to address complaints through dispute resolution. Complaints may be submitted to the AM at any time after concerns have



been brought directly to the attention of Bank Management and after Management has been given an opportunity to respond. For information on how to submit complaints to the Bank's GRS, visit <http://www.worldbank.org/GRS>. For information on how to submit complaints to the Bank's Accountability Mechanism, visit <https://accountability.worldbank.org>.

## VI. KEY RISKS

83. **The overall risk is assessed as High.** Noting the FCV context and resulting constant changes in the types and severities of project risks, a holistic approach to risk mitigation has been adopted through the MPA, including regular reassessment of the risk categories and mitigation measures on a bi-annual basis, incorporation of security and other risks into procurement documents, maintaining a geographically flexible approach to activity selection under future MPA Phases to allow incorporation of changes to the security and political environment, and systematically incorporating risk identification and mitigation into the MPA's learning agenda. Table 3 provides key risks and mitigation measures.

*Table 3: Project Risks and Mitigation*

| Risks                                                                                                                                                                                                                                                                                                                                                                     | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Political and Governance: High</b>                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Somalia's FCV context is highly complex, characterized by heightened political contestation and fragile intergovernmental relations, a weak social contract between the state and citizens, and deep-seated intercommunal divisions. Somalia continues to face protracted conflict and political brinksmanship, resulting in an uncertain political and security outlook. | <ul style="list-style-type: none"> <li>• Leveraging of MPA phases to iteratively scale-up and adapt based on lessons and FCV context, and encourage sustained engagement and learning</li> <li>• Program design leveraging consultation with all stakeholders and intergovernmental platforms</li> <li>• Geographic distribution of project sites to maximize flexibility</li> <li>• Evidence-based decision-making for the allocation of resources in future phases</li> <li>• Investments planned under Phase I benefitted from tailored FCV analysis, regular FCV and political economy support will be provided to identify and manage emerging risks in project implementation</li> </ul> |
| <b>Macroeconomics: Substantial</b>                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Somalia continues to be faced with low revenue mobilization, limited fiscal and monetary tools, and declining humanitarian and development assistance.                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>• Aims to strengthen Somalia's domestic revenue through reform of the aviation overflight fee structure as well as the overseeing governance structures.</li> <li>• Pillar 3 of the MPA will consistently strengthen the government's ability to capture and utilize sector revenue for operations and maintenance of assets.</li> </ul>                                                                                                                                                                                                                                                                                                                |
| <b>Sector Strategies and Policies: Substantial</b>                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| The infrastructure sector is challenged by unclear government roles in the infrastructure sector, including allocation of responsibility between levels of government and between ministries at the FGS level, associated weak coordination, and limited economic regulation on pricing.                                                                                  | <ul style="list-style-type: none"> <li>• A National Transport Policy is currently under development through the World Bank financed SHIP project, which will clearly delineate sectoral responsibilities. Other strategies, including a road safety strategy, are also under development and will guide implementation of the MPA.</li> <li>• The MPA's Pillar 2 will iteratively address sectoral challenges over time.</li> <li>• Aviation sector investments have been chosen with consideration for their potential to attract commercial operators and generate self-sustaining revenue.</li> </ul>                                                                                       |
| <b>Technical Design: Substantial</b>                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| The complexity of overall bundling of multi-sectoral investments in the MPA structure increases technical coordination demands.                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>• Project implementation structures are designed to ensure that implementing agencies have clear responsibility for activities within their mandates.</li> <li>• Project activities jointly identified by project stakeholders, while considering access constraints and readiness. More than 30 percent of project investments are ready for immediate tender at project approval.</li> </ul>                                                                                                                                                                                                                                                          |
| <b>Institutional capacity for implementation and sustainability: Substantial</b>                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |



|                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Challenges include fragmented institutions, dependence on international partners, shortages of qualified staff, and limited experience in some implementing entities.                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• Builds on a strong track record of delivery at the FGS and municipal level within the SURP-II and SHIP engagements, to be leveraged for delivery of the MPA</li> <li>• Where context and capacities encourage, the use of reputable implementing partners and UN, will ensure effective implementation and coordination.</li> </ul>                                                                                                                                                                                                                                           |
| <b>Fiduciary: High</b>                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Risks stem from nascent legal and operational systems and weak procurement practices. Prior procurements in Somalia highlight significant politization, delays, and integrity risk.                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• As with other World Bank projects, the MoF's External Assistance Fiduciary Department will be responsible for FM of the Project.</li> <li>• Ongoing PFM and transparency reforms at the country level are expected to further reduce residual risk for financial management</li> </ul>                                                                                                                                                                                                                                                                                        |
| <b>Environmental and Social: High</b>                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Environmental risks are driven by the program's scale, dispersion, and fragile context. MPA will fund multi-sector infrastructure across diverse ecological and social settings, increasing the chance of impacting fragile areas. Social risks include contextual, institutional, project factors that elevate likelihood of adverse impacts.       | <ul style="list-style-type: none"> <li>• Environmental and Social Impact Assessments, as well as Resettlement Action Plans are being developed for all investments and will be reviewed and cleared by the Bank prior to the beginning of project sites.</li> <li>• Aviation sector activities will be conducted entirely within the footprint of existing airports, requiring no land take.</li> <li>• Economic corridor activities have been designed to account for E&amp;S risk, including adjusting the proposed alignment to avoid environmental impacts.</li> </ul>                                             |
| <b>Stakeholders: High</b>                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Somalia is characterized by the lack of trust between key groups in government and society, and the accompanying high levels of contestation and violence, presenting a significant risk to all projects implemented in the country. In addition, the significant multi-sectoral coordination and decision-making required presents a risk of delay. | <ul style="list-style-type: none"> <li>• Individual project activities have been discussed with government, including through dissemination workshops during design process, incorporating feedback and local needs.</li> <li>• The proposed implementation arrangements have been designed to ensure relevant ministries maintain responsibility for delivery within their mandates.</li> <li>• Within the first Phase of the MPA all decisions on resource allocations were made prior to approval, preventing disputes between stakeholders.</li> </ul>                                                             |
| <b>Other – Security: High</b>                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Somalia continues to face significant insecurity challenges, violent extremist groups exerting control and influence across large parts of the country. Such challenges affect access to locations and security for staff and project stakeholders.                                                                                                  | <ul style="list-style-type: none"> <li>• Phase 1 activities have taken security risks, with prioritized activities in lower risk regions and/or lower exposure investments.</li> <li>• Security considerations are criteria for activity selection and prioritization</li> <li>• Mitigation includes close coordination with security authorities and local communities, use of Third-Party Monitoring, use of remote monitoring, and management of project visibility.</li> <li>• Regular engagement with the Government during project implementation to assess and manage the evolving security context.</li> </ul> |



## VII. RESULTS FRAMEWORK AND MONITORING

### Program Development Objective Indicators

#### Program Development Objective(s)

to increase access to resilient infrastructure and infrastructure sector jobs, and improve the capacity of select institutions.

#### PDO Indicators by PDO Outcomes

| Baseline                                                                                                                                         | Closing Period |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Increased Access to Resilient Infrastructure</b>                                                                                              |                |
| <b>Direct users that benefit from improved access to sustainable transport infrastructure and services (Number of people) <sup>CRI</sup></b>     |                |
| Apr/2026                                                                                                                                         | Jun/2031       |
| 0                                                                                                                                                | 820,000        |
| ➤ Direct users that benefit from improved access to sustainable transport infrastructure and services - Female (Number of people) <sup>CRI</sup> |                |
| Apr/2026                                                                                                                                         | Jun/2031       |
| 0                                                                                                                                                | 410,000        |
| ➤ Direct users that benefit from improved access to sustainable transport infrastructure and services - Youth (Number of people) <sup>CRI</sup>  |                |
| Apr/2026                                                                                                                                         | Jun/2031       |
| 0                                                                                                                                                | 574,000        |
| <b>People benefiting from climate resilient infrastructure (Number of people) <sup>CRI</sup></b>                                                 |                |
| Apr/2026                                                                                                                                         | Jun/2031       |
| 0                                                                                                                                                | 917,000        |
| ➤ People benefiting from climate resilient infrastructure - Female (Number of people) <sup>CRI</sup>                                             |                |
| Apr/2026                                                                                                                                         | Jun/2031       |
| 0                                                                                                                                                | 459,000        |
| ➤ People benefiting from climate resilient infrastructure - Youth (Number of people) <sup>CRI</sup>                                              |                |
| Apr/2026                                                                                                                                         | Jun/2031       |
| 0                                                                                                                                                | 642,000        |
| <b>Increased access to Infrastructure sector jobs</b>                                                                                            |                |
| <b>Direct person-days of employment generated by construction activities (Days)</b>                                                              |                |



|                                                                                                                                   |          |
|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| Apr/2026                                                                                                                          | Jun/2031 |
| 0                                                                                                                                 | 225,000  |
| ➤ Direct person-days of employment generated by construction activities - Female (Days)                                           |          |
| Apr/2026                                                                                                                          | Jun/2031 |
| 0                                                                                                                                 | 22,500   |
| ➤ Direct person-days of employment generated by construction activities - Youth (Days)                                            |          |
| Apr/2026                                                                                                                          | Jun/2031 |
| 0                                                                                                                                 | 157,500  |
| <b>Improved Capacity of Select Institutions</b>                                                                                   |          |
| <b>Government staff receiving training in core technical disciplines and receiving required licenses (as applicable) (Number)</b> |          |
| Apr/2026                                                                                                                          | Jun/2031 |
| 0                                                                                                                                 | 40       |

#### Intermediate Indicators by Components

| Baseline                                                                                         | Closing Period |
|--------------------------------------------------------------------------------------------------|----------------|
| <b>Resilient Infrastructure for Connectivity, Jobs, and Growth</b>                               |                |
| <b>Airports upgraded to a ICAO Code 3C Classification (Number)</b>                               |                |
| Apr/2026                                                                                         | Jun/2031       |
| 0                                                                                                | 2              |
| ➤ Airfield runways rehabilitated or maintained with climate-resilient measures (Number) (Number) |                |
| Apr/2026                                                                                         | Jun/2031       |
| 0                                                                                                | 2              |
| <b>Installation of Airfield Ground Lighting for international diversion access (Number)</b>      |                |
| Apr/2026                                                                                         | Jun/2031       |
| 0                                                                                                | 2              |
| <b>Airfield rescue and firefighting (ARFF) vehicles installed in airports (Number)</b>           |                |
| Apr/2026                                                                                         | Jun/2031       |
| 0                                                                                                | 9              |
| <b>Length of drainage constructed or rehabilitated (Kilometers)</b>                              |                |
| Apr/2026                                                                                         | Jun/2031       |
| 0                                                                                                | 58.87          |
| ➤ Length of trunk drainage constructed (Kilometers)                                              |                |
| Apr/2026                                                                                         | Jun/2031       |



|                                                                                                                                                                           |          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 0                                                                                                                                                                         | 2.87     |
| <b>&gt;Length of ancillary drainage constructed or rehabilitated (Kilometers)</b>                                                                                         |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| 0                                                                                                                                                                         | 56       |
| <b>Length of economic corridor constructed (Kilometers)</b>                                                                                                               |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| 0                                                                                                                                                                         | 63       |
| <b>Percentage of sampled beneficiaries satisfied that infrastructure met their needs (Percentage)</b>                                                                     |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| 0                                                                                                                                                                         | 80       |
| <b>Percentage of women in project areas who consider that their needs and priorities have been taken into account in final planning decisions and design (Percentage)</b> |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| 0                                                                                                                                                                         | 65       |
| <b>Institutional Transformation and Human Capacity Building</b>                                                                                                           |          |
| <b>Roadmap for implementation of Somali Aviation Authority developed (Yes/No)</b>                                                                                         |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| No                                                                                                                                                                        | Yes      |
| <b>Roadmap for implementation of Somali engineer's board developed (Yes/No)</b>                                                                                           |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| No                                                                                                                                                                        | Yes      |
| <b>Air Navigation Service professionals employed by the Somalia Civil Aviation Authority who hold the required licenses (Percentage)</b>                                  |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| 17                                                                                                                                                                        | 41       |
| <b>New fully licensed Air Traffic Management (ATM) professionals employed by SCAA who are women (Number)</b>                                                              |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| 0                                                                                                                                                                         | 10       |
| <b>Sustainable Operations and Management</b>                                                                                                                              |          |
| <b>Flight Information Region coverage and reliability provided by upgraded air navigation systems (Percentage)</b>                                                        |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| 0                                                                                                                                                                         | 100      |
| <b>Improved communication, navigation and surveillance systems (Yes/No)</b>                                                                                               |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |
| No                                                                                                                                                                        | Yes      |
| <b>&gt;Installation of Automatic Dependent Surveillance – Broadcast (ADS-B) ground stations (Number)</b>                                                                  |          |
| Apr/2026                                                                                                                                                                  | Jun/2031 |



|                                                                                                                         |          |
|-------------------------------------------------------------------------------------------------------------------------|----------|
| 0                                                                                                                       | 7        |
| ➤ Installation of Very Small Aperture Terminal (VSAT) network (Number)                                                  |          |
| Apr/2026                                                                                                                | Jun/2031 |
| 0                                                                                                                       | 7        |
| Share of project investments with costed operations and maintenance plans adopted by responsible authority (Percentage) |          |
| Apr/2026                                                                                                                | Jun/2031 |
| 0                                                                                                                       | 100      |
| Infrastructure Implementation Coordination and Management                                                               |          |
| Roadmap developed for transition to use of civil servants in project implementation (Yes/No)                            |          |
| Apr/2026                                                                                                                | Jun/2031 |
| No                                                                                                                      | Yes      |



## Monitoring & Evaluation Plan: PDO Indicators by PDO Outcomes

| <b>Increased Access to Resilient Infrastructure</b>                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Direct users that benefit from improved access to sustainable transport infrastructure and services (Number of people) <sup>CRI</sup></b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                                                                           | The Corporate Results Indicator counts the number of people who directly use, or can reliably access, transport assets and services upgraded or provided by the Project. Direct users are people living or working within the functional catchment of an improved transport asset, as well as regular users of services. In the case of this project, includes the direct users of the economic corridor, as well as the project airports. Due to the nature of the Somali air network and Mogadishu's role as the central hub, AAIA passengers is used as a proxy for overall passengers to avoid double counting. For specific methodological guidance, see the WBG Scorecard Results indicator methodology.                                                                                                                                                                                                                                                                                                                                                                                              |
| Frequency                                                                                                                                             | Project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data source                                                                                                                                           | SCAA air traffic monitoring; road corridor traffic counts at project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Methodology for Data Collection                                                                                                                       | Aviation data based on regular monitoring by SCAA. Road corridor data to measured through a traffic count at project close.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Responsibility for Data Collection                                                                                                                    | MoTCA, MoPWRH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Direct users that benefit from improved access to sustainable transport infrastructure and services - Female (Number of people) <sup>CRI</sup></b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                                                                           | Sub-indicator to measure the number of female beneficiaries – calculated as 50% of total beneficiaries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Frequency                                                                                                                                             | Project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data source                                                                                                                                           | SCAA air traffic monitoring; road corridor traffic counts at project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Methodology for Data Collection                                                                                                                       | 50 percent of parent indicator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Responsibility for Data Collection                                                                                                                    | MoTCA, MoPWRH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Direct users that benefit from improved access to sustainable transport infrastructure and services - Youth (Number of people) <sup>CRI</sup></b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                                                                           | Sub-indicator to measure the number of youth beneficiaries – calculated as 70% percent of total beneficiaries based on census data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Frequency                                                                                                                                             | Project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data source                                                                                                                                           | SCAA air traffic monitoring; road corridor traffic counts at project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Methodology for Data Collection                                                                                                                       | 70 percent of parent indicator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Responsibility for Data Collection                                                                                                                    | MoTCA, MoPWRH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>People benefiting from climate resilient infrastructure (Number of people) <sup>CRI</sup></b>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                                                                           | The Corporate Results Indicator counts the number of people who benefit directly or indirectly from investments that integrate climate-resilient/adaptation measures into infrastructure and the services it provides (e.g., roads and bridges with all-weather access, flood protection, drainage, water/sanitation, shelters, nature-based infrastructure). Benefits may include reduced service disruption and damage from climate hazards, year-round connectivity, improved safety and access to services, and reduced exposure to floods and other climate risks. In this project, it is measured as the sum of the direct and indirect road corridor users, calculated as per the corporate results indicator methodology for “Millions of people that benefit from improved access to sustainable transport infrastructure and services”; the beneficiaries of the Mogadishu trunk drainage improvements calculated using a 250 meter buffer to project investments (utilizing WorldPop population distribution); and the airport passengers (proxied by annual passengers traveling through AAIA). |
| Frequency                                                                                                                                             | Project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data source                                                                                                                                           | SCAA air traffic monitoring; road corridor traffic counts at project close; WorldPop                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Methodology for Data Collection                                                                                                                       | Assessed individually for each investment in line with the above methodologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Responsibility for Data Collection                                                                                                                    | MoTCA, MoPWRH, BRA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>People benefiting from climate resilient infrastructure - Female (Number of people) <sup>CRI</sup></b>                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                                                                           | Sub-indicator to measure the number of female beneficiaries – calculated as 50% of total beneficiaries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Frequency                                                                                                                                             | Project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



|                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data source                                                                                                              | SCAA air traffic monitoring; road corridor traffic counts at project close                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Methodology for Data Collection                                                                                          | 50 percent of parent indicator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Responsibility for Data Collection                                                                                       | MoTCA, MoPWRH, BRA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>People benefiting from climate resilient infrastructure - Youth (Number of people) <sup>CRI</sup></b>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description                                                                                                              | Sub-indicator to measure the number of youth beneficiaries – calculated as 70% of total beneficiaries                                                                                                                                                                                                                                                                                                                                                                                                            |
| Frequency                                                                                                                | Project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Data source                                                                                                              | SCAA air traffic monitoring; road corridor traffic counts at project close                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Methodology for Data Collection                                                                                          | 70 percent of parent indicator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Responsibility for Data Collection                                                                                       | MoTCA, MoPWRH, BRA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Increased Access to Infrastructure Sector Jobs</b>                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Direct person-days of employment generated by construction activities (Days) <sup>CRI</sup></b>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description                                                                                                              | The Corporate Results Indicator (more and better jobs) counts the number of people who benefit directly from employment opportunities through public works programs. Workers directly engaged in construction for the Project (prime contractor and subcontractors), including site management, skilled and unskilled labor, safety staff, and quality control. Off-site workers in facilities dedicated to the Project (e.g., concrete batching, pre-cast yards) where ≥75% of their output is for the Project. |
| Frequency                                                                                                                | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data source                                                                                                              | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports; Contractor payroll records                                                                                                                                                                                                                                                                                                                                                                         |
| Methodology for Data Collection                                                                                          | Sum of days worked by all eligible workers during the period                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Responsibility for Data Collection                                                                                       | MoTCA, MoPWRH, BRA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Direct person-days of employment generated by construction activities (Days) - Female <sup>CRI</sup></b>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description                                                                                                              | Sub-indicator to measure the number of female beneficiaries                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Frequency                                                                                                                | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data source                                                                                                              | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports; Contractor payroll records                                                                                                                                                                                                                                                                                                                                                                         |
| Methodology for Data Collection                                                                                          | Sum of days worked by all eligible workers during the period                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Responsibility for Data Collection                                                                                       | MOTCA, MoPWRH, BRA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Direct person-days of employment generated by construction activities (Days) – Youth <sup>CRI</sup></b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description                                                                                                              | Sub-indicator to measure the number of youth beneficiaries                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Frequency                                                                                                                | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data source                                                                                                              | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports; Contractor payroll records                                                                                                                                                                                                                                                                                                                                                                         |
| Methodology for Data Collection                                                                                          | Sum of days worked by all eligible workers during the period                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Responsibility for Data Collection                                                                                       | MoTCA, MoPWRH, BRA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Improved Capacity of Select Institutions</b>                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Government staff receiving training in core technical disciplines and receiving required licenses (as applicable)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description                                                                                                              | Count of the number of people who are trained in their core technical disciplines within the implementing Ministries. To include Air Navigation Specialists within MoTCA (and SCAA), engineers within MoPWRH and other technical staff. For ANS Specialists, will include only those whose training results in receiving an additional rating for their license. For engineers, any training on engineering will be counted, pending the establishment of a Somali engineers' Board                              |
| Frequency                                                                                                                | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data source                                                                                                              | Project Implementation Unit Monitoring Reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



|                                    |                                                                               |
|------------------------------------|-------------------------------------------------------------------------------|
| Methodology for Data Collection    | Reporting by implementing agencies on the Project supported training outcomes |
| Responsibility for Data Collection | MoTCA, MoPWRH, MoF                                                            |

## Monitoring & Evaluation Plan: Intermediate Results Indicators by Components

| Resilient Infrastructure for Connectivity, Jobs, and Growth                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Airports upgraded to a ICAO Code 3C Classification                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Description                                                                           | The indicator measures the number of project airports which meet an ICAO 3C classification based on project investments                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Frequency                                                                             | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Data source                                                                           | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports. SCAA inspection/audit reports                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Methodology for Data Collection                                                       | Assessment of airfield designs; SCAA certification of airport                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Responsibility for Data Collection                                                    | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Airfield runways rehabilitated or maintained with climate-resilient measures (Number) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Description                                                                           | The indicator measures the number of runway pavements and/or airstrips at domestic or international airfields that have been rehabilitated or received scheduled maintenance and integrates defined climate resilient measures aligned with the Project's engineering specifications.                                                                                                                                                                                                                                                                                         |
| Frequency                                                                             | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Data source                                                                           | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports. SCAA inspection/audit reports                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Methodology for Data Collection                                                       | Assessment of airfield designs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Responsibility for Data Collection                                                    | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Installation of Airfield Ground Lighting for international diversion access (Number)  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Description                                                                           | The indicator measures the number of aerodromes that have installed and commissioned airfield ground lighting systems enabling safe night/low-visibility operations for international diversion, in accordance with ICAO Annex 14 and applicable national standards. "AGL system" includes runway edge lights, threshold/end lights, runway end identifier lights (as applicable), approach lighting system appropriate to runway category, PAPI/VASI, taxiway lights or reflectors sufficient for safe movement, AGL power and control (CCR, regulators), and standby power. |
| Frequency                                                                             | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Data source                                                                           | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports;                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Methodology for Data Collection                                                       | Engineer's SAT/commissioning certificates; and SCAA inspection/audit reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Responsibility for Data Collection                                                    | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Airfield rescue and firefighting (ARFF) vehicles (Number)                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Description                                                                           | The indicator counts the number of specialized ARFF vehicles procured, delivered, and commissioned into operational service at designated aerodromes in accordance with ICAO Annex 14 and national requirements for each project site's ARFF category.                                                                                                                                                                                                                                                                                                                        |
| Frequency                                                                             | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Data source                                                                           | SAT certificates; registration/insurance documents; inventory of agents and equipment; crew training records; operational roster entries.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Methodology for Data Collection                                                       | Project Implementation Unit Monitoring Reports; SCAA inspection/audit reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Responsibility for Data Collection                                                    | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



| <b>Length of drainage constructed or rehabilitated (Kilometers)</b>                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                                                                           | Measures the total centerline length, in kilometers, of drainage infrastructure that has been newly constructed or rehabilitated to restore design hydraulic performance and structural integrity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Frequency                                                                                             | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Data source                                                                                           | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Methodology for Data Collection                                                                       | Review of supervising engineer progress reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Responsibility for Data Collection                                                                    | MoPWRH, MoTCA, BRA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Length of trunk drainage constructed (Kilometers)</b>                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                           | Measures the total centerline length, in kilometers, of trunk (primary) drainage infrastructure that has been newly constructed or rehabilitated to restore design hydraulic performance and structural integrity. "Trunk drainage" refers to primary/major conduits that convey runoff to outfalls or retention/detention systems (e.g., main open channels, box/pipe culverts of primary networks, large storm sewers). It may include secondary collectors where they function as part of the backbone network serving sizable catchments. Drainage assets minimum size thresholds (e.g., ≥1.0 m bottom width for open channels, ≥1.2 m diameter or box equivalent for culverts/storm sewers), adapted to local standards and hydrology. |
| Frequency                                                                                             | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Data source                                                                                           | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Methodology for Data Collection                                                                       | Review of supervising engineer progress reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Responsibility for Data Collection                                                                    | MoPWRH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Length of ancillary drainage constructed or rehabilitated (Kilometers)</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                           | Measures the total centerline length, in kilometers, of ancillary drainage infrastructure that has been newly constructed or rehabilitated to restore design hydraulic performance and structural integrity. In the Project, "Ancillary drainage" specifically refers to longitudinal drainage along roadways and runways.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Frequency                                                                                             | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Data source                                                                                           | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Methodology for Data Collection                                                                       | Review of supervising engineer progress reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Responsibility for Data Collection                                                                    | MoPWRH, MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Length of economic corridor constructed (Kilometers)</b>                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                           | Measures the total centerline length, in kilometers, of roadway built in the Project which serves primary trade and transit traffic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Frequency                                                                                             | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Data source                                                                                           | Project Implementation Unit Monitoring Reports; Contractor and supervising engineer monthly progress reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Methodology for Data Collection                                                                       | Review of supervising engineer progress reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Responsibility for Data Collection                                                                    | MoPWRH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Percentage of sampled beneficiaries satisfied that infrastructure met their needs (Percentage)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                                                                                           | This indicator measures the satisfaction levels of beneficiaries who directly benefit from improved roads and infrastructure under the Project. Satisfaction will be assessed through focus group discussions (FGDs) and/or surveys, evaluating multiple dimensions, including Quality of rehabilitated infrastructure and Improved access to services and jobs.                                                                                                                                                                                                                                                                                                                                                                            |
| Frequency                                                                                             | Project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data source                                                                                           | FGDs conducted as part of project completion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Methodology for Data Collection                                                                       | Structured focus group discussions held during the Project's completion phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Responsibility for Data Collection                                                                    | MoPWRH, MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



|                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Collection                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Percentage of women in project areas who consider that their needs and priorities have been taken into account in final planning decisions and design (Percentage)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                                                                                                                                                               | This indicator measures the extent of meaningful participation of women in project implementation by assessing whether they feel their needs and priorities were reflected in final planning and design decisions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Frequency                                                                                                                                                                 | Project close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Data source                                                                                                                                                               | FDGs conducted as part of project completion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Methodology for Data Collection                                                                                                                                           | Structured focus group discussions held during the Project's completion phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Responsibility for Data Collection                                                                                                                                        | MoPWRH, MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Institutional Transformation and Human Capacity Building</b>                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Roadmap for implementation of Somali Aviation Authority (Yes/No)</b>                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                                                                                                                                                               | A comprehensive roadmap document that sets out the phased institutional, legal, operational, financial, and HR steps to establish and operationalize the Somali Aviation Authority (SAA), including timelines, responsibilities, resource needs, risk management, and a monitoring framework, is prepared and formally endorsed by the competent authority.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Frequency                                                                                                                                                                 | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Data source                                                                                                                                                               | Project Implementation Unit Monitoring Reports;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Methodology for Data Collection                                                                                                                                           | Consultant final deliverables accepted by MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Responsibility for Data Collection                                                                                                                                        | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Roadmap for implementation of Somali engineers' board developed (Yes/No)</b>                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                                                                                                                                                               | A comprehensive roadmap document that sets out the phased institutional, legal, operational, financial, and HR steps to establish and operationalize the Somali engineers' board, including timelines, responsibilities, resource needs, risk management, and a monitoring framework, is prepared and formally endorsed by the competent authority.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Frequency                                                                                                                                                                 | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Data source                                                                                                                                                               | Project Implementation Unit Monitoring Reports;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Methodology for Data Collection                                                                                                                                           | Consultant final deliverables accepted by MoPWRH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Responsibility for Data Collection                                                                                                                                        | MoPWRH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Air Navigation Service professionals employed by the Somalia Civil Aviation Authority who hold the required licenses (percentage)</b>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                                                                                                                                                               | The indicator measures the share of approved technical/operational aviation positions at the Somalia Civil Aviation Authority (SCAA) that are filled by qualified personnel. Licensed aviation professionals are staff in safety- and service-critical cadres such as: air traffic services, air navigation services (CNS engineering), aeronautical information management, flight operations and airworthiness inspectors, aerodrome safety/standards, airfield rescue and firefighting, search and rescue coordination, aviation security (as per national regulations), and meteorology liaison posted to aviation operations. Qualified means the employee meets the minimum requirements for the post under the State's regulations (e.g., valid license/endorsement where applicable, required recurrent training and medical exams). |
| Frequency                                                                                                                                                                 | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Data source                                                                                                                                                               | Project Implementation Unit Monitoring Reports; Licensing and training records (SCAA licensing unit, inspectorate files) to verify that filled posts meet qualification requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Methodology for Data Collection                                                                                                                                           | SCAA approved organizational structure and annual staffing plan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Responsibility for Data Collection                                                                                                                                        | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>New fully licensed Air Traffic Management (ATM) professionals employed by SCAA who are women (number)</b>                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                                                                                                                                                               | This indicator measures the number of new women who, through the Project, successfully obtain a full Air Traffic Management (ATM) operational license and are employed by SCAA. ATM refers to the operational functions responsible for managing aircraft movements safely and efficiently, including Air Traffic Control (ATC), Air Traffic Flow Management (ATFM), and Airspace Management (ASM).                                                                                                                                                                                                                                                                                                                                                                                                                                          |



|                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                    | <p>“Fully licensed” means that the individual meets the operational requirements under national and international aviation regulations, including valid licensing/endorsements, required recurrent training, and medical certification, and is operationally deployable. Achieving full licensing requires completion of a structured sequence of advanced training and certification programs, including Area Control Procedural, Approach Control Procedural, Aerodrome Control (ICAO 052), On-the-Job Training Instructor Course, and CPDLC/ADS-B systems training.</p> <p>Baseline: The baseline is 0, as the indicator measures new fully licensed women supported by the Project. While SCAA currently has 86 ATM professionals (75 men and 11 women), of which 15 are fully licensed (14 men and 1 woman, i.e., 7% women among fully licensed staff), the existing fully licensed woman is not a result of project-supported training and is therefore not counted in the baseline.</p> <p>Project target and expected results: Considering the cost and duration of training and certification programs, the MoTCA proposes to fully license 20 additional ATM professionals through the Project, of which 10 are expected to be women. By the end of the Project, the total number of fully licensed ATM professionals is expected to increase from 15 to 35, and the number of fully licensed women from 1 to 11, increasing their share among fully licensed staff from approximately 7% to 30%.</p> |
| Frequency                                                                                                          | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data source                                                                                                        | Project Implementation Unit Monitoring Reports; Licensing and training records (SCAA licensing unit, inspectorate files) to verify that filled posts meet qualification requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Methodology for Data Collection                                                                                    | SCAA approved organizational structure and annual staffing plan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Responsibility for Data Collection                                                                                 | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Sustainable Operations and Management</b>                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Flight Information Region coverage and reliability provided by upgraded air navigation systems (Percentage)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description                                                                                                        | The indicator measures the proportion of the Somali Flight Information region airspace where upgraded air navigation systems (e.g., surveillance, CNS/ATM, VHF/HF/CPDLC communications, navigation aids, ADSB) provide service at or above the prescribed performance targets (e.g., availability, accuracy, integrity) for applicable flight levels/routes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Frequency                                                                                                          | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data source                                                                                                        | Project Implementation Unit Monitoring Reports; System performance logs (availability/uptime), NAVAID monitoring systems, ADS-B/radar track coverage reports, VHF coverage surveys, ATM incident logs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Methodology for Data Collection                                                                                    | Establish the baseline via pre-upgrade coverage maps and historical availability logs (e.g., AFTN/ATM system logs, SNMP/SCADA, maintenance records) for the 12 months prior to upgrades.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Responsibility for Data Collection                                                                                 | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Improved communication, navigation and surveillance systems (Yes/No)</b>                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description                                                                                                        | The yes/no indicator is a composite of the installation of (a) Automatic Dependent Surveillance – Broadcast (ADS-B) ground stations (ii) Very Small Aperture Terminal (VSAT) network; and (iii) Instrument Landing System for MGQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Frequency                                                                                                          | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data source                                                                                                        | SAT certificates and SDP/NMS activation logs; validated by supervision consultant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Methodology for Data Collection                                                                                    | Civil works completed, equipment installed and configured, timing synchronized, data link established to SDP, Site Acceptance Testing (SAT) passed, and the site is visible and health-monitored in the Network/Surveillance Management System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Responsibility for Data Collection                                                                                 | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Installation of Automatic Dependent Surveillance – Broadcast (ADS-B) ground stations (Number)</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description                                                                                                        | The number of ADS-B ground stations installed and commissioned into the national surveillance network in accordance with approved technical specifications, providing validated reception and backhaul of 1090ES ADS-B messages to the surveillance data processor/ATM system. An ADS-B ground station includes antenna(s), receiver(s), RF front-end, GPS/GNSS timing, processing unit, power and grounding, shelter/rack, and secure data backhaul integrated with the Surveillance Data Processor (SDP) or equivalent system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Frequency                                                                                                          | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data source                                                                                                        | SAT certificates and SDP/NMS activation logs; validated by supervision consultant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Methodology for Data                                                                                               | Civil works completed, equipment installed and configured, timing synchronized, data link established to SDP, Site                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



|                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Collection                                                                                                                     | Acceptance Testing (SAT) passed, and the site is visible and health-monitored in the Network/Surveillance Management System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Responsibility for Data Collection                                                                                             | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Installation of Very Small Aperture Terminal (VSAT) network (Number)</b>                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description                                                                                                                    | The number of distinct VSAT ground stations (sites) installed and commissioned into the national ANSP/CAA communications network in accordance with the approved technical specifications, providing operational connectivity for air navigation or aeronautical administration. A "VSAT site" includes antenna, RF chain (BUC/LNB/modem), indoor unit, power and grounding, shelters/racks, and network integration with the hub/NMS. "Installed and commissioned" means civil works completed, equipment installed and configured, link established to designated satellite/hub, acceptance tests passed, and site entered into the Network Management System (NMS) with an assigned node ID. |
| Frequency                                                                                                                      | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data source                                                                                                                    | Engineer's SAT certificates; hub/NMS node activation records; link availability test reports; as-built documentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Methodology for Data Collection                                                                                                | Civil works completed, equipment installed and configured, timing synchronized, data link established to SDP, Site Acceptance Testing (SAT) passed, and the site is visible and health-monitored in the Network/Surveillance Management System                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Responsibility for Data Collection                                                                                             | MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Share of project investments with costed operations and maintenance plans adopted by responsible authority (Percentage)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description                                                                                                                    | The number of project investments for which O&M plans have been developed which include costing. For measurement of this indicator, project investments will be counted by discrete project sites (each airport, road, drainage system). The indicator will be considered met when the responsible authority for the specific investment formally adopts the operations and maintenance plan.                                                                                                                                                                                                                                                                                                   |
| Frequency                                                                                                                      | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data source                                                                                                                    | Project Implementation Unit Monitoring Reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Methodology for Data Collection                                                                                                | Receipt of communication from responsible authority                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Responsibility for Data Collection                                                                                             | MoPWRH, MoTCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Sustainable Operations and Management</b>                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Roadmap developed for transition to use of civil servants in project implementation (Yes/No)</b>                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description                                                                                                                    | This indicator measures the achievement in the preparation of a roadmap for transition to the use of civil servants in project implementation. This indicator will be considered met when the roadmap has been accepted by the Ministry of Finance (as implementing agency of the relevant activity)                                                                                                                                                                                                                                                                                                                                                                                            |
| Frequency                                                                                                                      | Bi-annual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data source                                                                                                                    | Project Monitoring Reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Methodology for Data Collection                                                                                                | Receipt of communication from responsible authority                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Responsibility for Data Collection                                                                                             | MoF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



## ANNEX 1: Implementation Arrangements and Support Plan

### COUNTRY: Somalia, Federal Republic of Somalia Infrastructure Development for Access and Jobs, Phase 1

1. **The Project will have three implementing agencies: the MoF, MoTCA and the MoPWRH.** Teams from these Ministries, jointly as a National MPA Coordination Unit, will be responsible for the overall implementation of all activities in the Project. Implementation responsibilities by Ministry include:

- **Ministry of Finance (MoF)** – The MoF is responsible for overall coordination of the MPA, as well as implementation of Sub-Component 2.C. Designated project consultants reporting to the MoF will include a Program Coordinator, an M&E/Communications Specialist, a Procurement Specialist, and a Financial Management Specialist. A dedicated implementation budget will be established as part of Component 4 to support the MoF implementation costs.
- **Ministry of Public Works, Reconstruction and Housing (MoPWRH)** - For activities in Sub-Components 1.B and 1.C, the existing SURP-II and SHIP Project Coordination Units under the MoPWRH will take the technical lead in implementation of urban drainage and economic corridors, respectively, through the closure of these projects. At that point, the two teams will be consolidated into a unified Project Implementation Unit within the Ministry. Sub-components 2.B and relevant activities in 3.B will also be led by the MoPWRH. The MoPWRH team will be responsible for the procurement, contract management, financial management, E&S risk management, and other implementation aspects of these activities. For activities in Sub-component 1.B, which will be directly implemented by the BRA, MoPWRH will provide oversight and technical backstopping. For activities in Sub-Component 1.C, the MoPWRH will engage a UN partner to deliver activities in Somaliland. A dedicated implementation budget will be established as part of Component 4 to support the MoPWRH implementation costs.
- **Ministry of Transport and Civil Aviation (MoTCA)**- Air transport activities (including Sub-Components 1.A, 2.A, 3.A and relevant activities in 3.B) will be led by a Project Implementation Unit to be established within the MoTCA, including a Project Coordinator, a Procurement Specialist, E&S specialists, a Financial Management Specialist, an Airports Engineer, and an Air Navigation System Specialist. A dedicated implementation budget will be established as part of Component 4 to support MoTCA implementation costs.

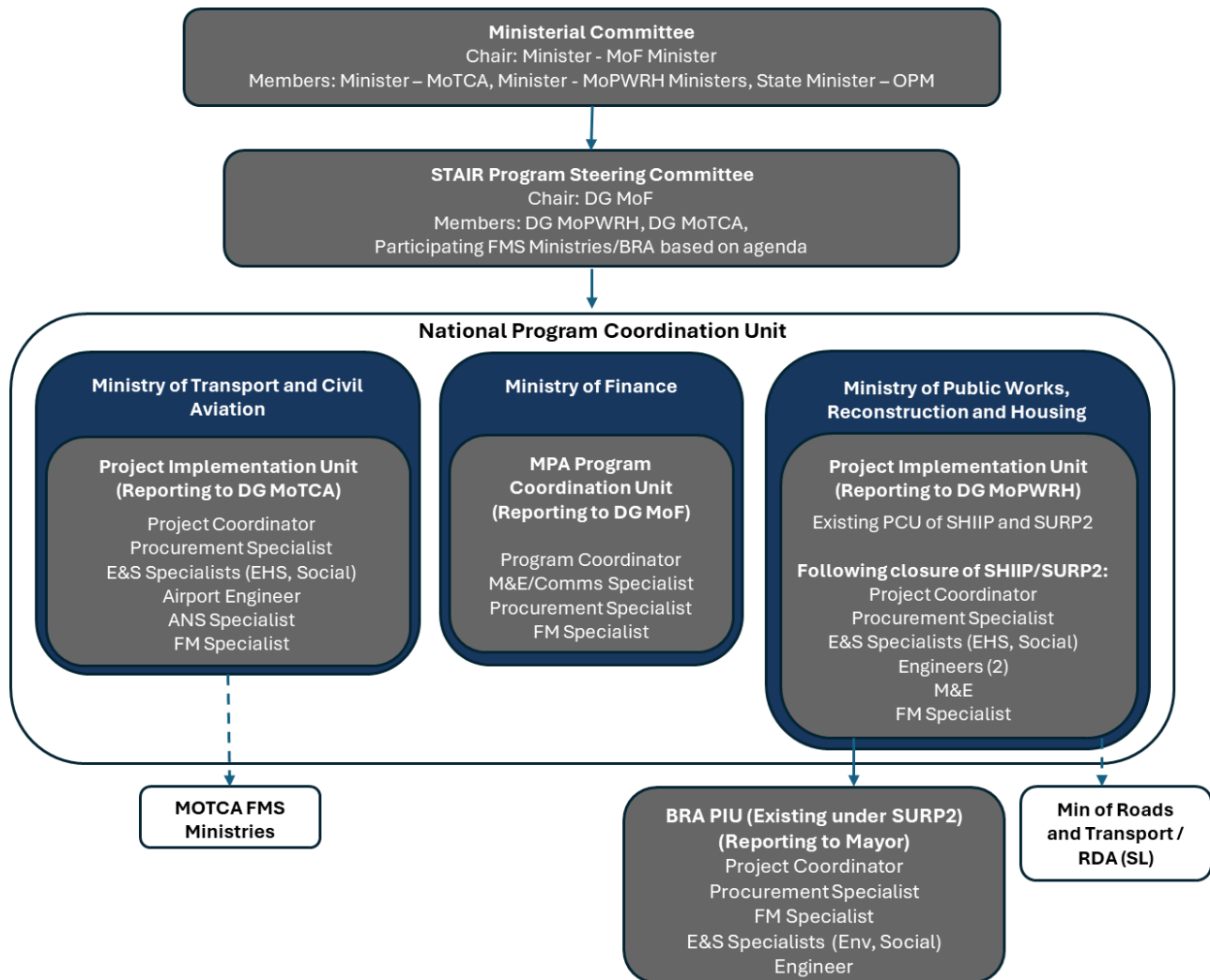
2. **Project governance will be led by two committees: a Ministerial Committee and a Technical Steering Committee.** The Ministerial Committee, chaired by the Minister of Finance and including the Ministers of MoTCA and MoPWRH, and the State Minister of the Office of the Prime Minister, will meet on a bi-annual basis to review the overall MPA program progress and set high level direction for ongoing and future phases. Day-to-day governance will be the responsibility of a Technical Steering Committee led by the DG of the MoF and consisting of the DGs of the MoTCA and the MoPWRH, as well as representatives from engaged FMS ministries/BRA depending on the agenda, which will meet on a quarterly basis, or more often as required.

3. **The Project will leverage existing capacities in the MoF, the MoPWRH, and city governments, while building on a longstanding engagement with the MoTCA.** For urban drainage activities, the BRA Project Implementation Unit established under the SURP-II project will be leveraged. Where airport project sites align (in Mogadishu, Baidoa, Kismayo and Garowe), city level PIUs already in place under the SURP-II project may be leveraged to support implementation as relevant. Clear tri-partite Memorandums of Understanding will be



established between FMS and FGS to define roles, responsibilities, and coordination mechanisms, which will be reflected in the Project operations manual. The Project will move away from stand-alone project structures and enable a gradual transition to greater reliance on civil servants, supported by upfront transition planning during Phase 1 to strengthen long-term sustainability and capacity. See below for further details on implementation arrangements.

*Stairs2Jobs Phase 1 Organogram*





## ANNEX 2: Project Map

